HEIRSTONE CONSULTING

HEIRSTONE CONSULTING RESEARCH | 2026

CONSTRUCTION COSTS AND BUILDING TECHNOLOGY

Nine cities, one square metre

What it costs to build in nine African cities, why the costs differ, and when building technology brings them down, with Cairo as the comparator



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THE REPORT IN SIX NUMBERS

2.9x

AECOM's price for a multi-storey residential block in Lagos against Nairobi on the same 2024/25 guide: US$2,330 against US$796 per m².

US$4.13

a 50 kg bag of cement in Egypt in 2025, the lowest of the ten cities. The same bag cost US$11.03 in Kinshasa.

14%

the weight of cement and lime in Kenya's building cost index. Labour weighs 21% and steel 11%.

5% to 39%

interest on a two-year, fully debt-funded build as a share of hard cost, from Casablanca to Lagos.

57 to 59%

added to the price of imported cement by duty and VAT in Kenya and Tanzania, against 14% in Egypt.

15%

the saving reported on a completed 3D-printed school in Malawi. Dubai's 3D printing strategy had stated 50 to 70%.

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THE CITIES

Kinshasa (DR Congo), Douala (Cameroon), Libreville (Gabon), Lagos (Nigeria), Abidjan (Côte d'Ivoire), Dakar (Senegal), Nairobi (Kenya), Dar es Salaam (Tanzania) and Casablanca (Morocco). Comparator: Cairo (Egypt). References: Accra, Johannesburg, Dubai, Riyadh and London.

ABOUT THIS REPORT

This report compares what it costs to build a square metre in nine African cities, with Cairo as the regional comparator and Gulf and European cities as reference points. It splits the gap between them into materials, duties and taxes, labour, contractors' overheads and profit, haulage, and the cost of money and time, and it tests whether industrialised building lowers cost or time.

It is written for developers and contractors, development finance institutions, lenders and investors, corporate occupiers planning offices and warehouses, and the ministries and agencies that set duties, codes and housing programmes. It draws on public sources current in 2026: cost consultants' guides, the Centre for Affordable Housing Finance in Africa, central banks, statistics offices, tariff schedules, company reports and the trade press.

> **Reading the numbers.** Costs are US$ per m² of gross floor area, excluding land, fees, VAT and finance unless stated. Local-currency figures are converted at the rate each source used, and the year of that rate is given. Figures we derive are labelled Heirstone analysis, with their inputs.

WHAT WE FOUND

The nine cities do not share one cost problem. In each, a different layer of the square metre is the expensive one, and building technology is rarely the first fix.

We assembled published construction costs, cement and steel prices, tariffs, lending rates, wages and the evidence on industrialised building for nine cities and Cairo into the Heirstone Build Cost Index 2026. We then measured how many points of construction cost each lever could save in each city.

1 **Published costs span three to one, and six of the nine cities have no current guide rate.**

AECOM prices a multi-storey residential block at US$2,330 per m^2 in Lagos, US$893 in Dar es Salaam and US$796 in Nairobi. Kinshasa, Douala, Libreville, Abidjan and Dakar have only national or regional averages, mostly prestige-office figures, and Casablanca has no published construction cost. Egyptian apartments, at US$720 to 1,270 per m^2 to local specification, cost 45 to 69% less than Lagos.

2 **Cement is a smaller lever than its price suggests.**

A bag costs US$4.13 in Egypt and US$11.03 in Kinshasa, yet cement and lime weigh 14% in Kenya′s building cost index. Nigeria′s competition regulator puts capacity at 60 to 65 Mt for 25 to 30 Mt of demand, yet bags cost US$135 to 217 a tonne in 2026. Egypt runs its kilns at about 86% of capacity and sold at US$101 to 106 a tonne even at the 2025 peak.

3 **The border and the road add as much as 60% to a bag of cement.**

Duty and VAT add 57 to 59% to imported cement in Kenya and Tanzania, 41 to 43% in the CFA franc zones and Morocco, and 14% in Egypt. Gabon′s official price list puts cement 37 to 60% higher in the interior than around Libreville, and Kinshasa′s price halved from US$21 to US$11 a bag when a transport strike ended.

4 **In the dearest cities, money and margin outweigh materials.**

Interest on a two-year build is 5% of hard cost in Casablanca and 21 to 39% in Lagos, where contractors also add 35% for overheads and profit on a small job against 15% in Nairobi. Cameroon and Gabon share a 4.50% policy rate, yet bank credit cost 9.03% and 21.51% respectively.

5 **Technology pays through time, and only at volume.**

Claimed savings fall from 50 to 70% in Dubai′s strategy to 10 to 15% on completed projects. Time savings of 30 to 50% hold up better, and they are worth most where money is dear: taking a quarter off a two-year programme saves 6 to 12 points of hard cost in Lagos and about 1.6 in Dakar.

WHAT IT MEANS

Governments should start with the lever that is largest in their own city: duty in East Africa, credit in Lagos and Libreville, clinker and haulage in Central Africa. They should also publish a construction cost index.

Lenders and DFIs can cut more cost with long-tenor construction finance than most technologies can.

Developers should package work at scale, source structure regionally and back a factory only when it passes the Heirstone Volume Test.

10

cities in the Heirstone Build Cost Index 2026, with Cairo as comparator

5

rungs on the Heirstone Cost Ladder: speed, credit, margin, border and cement

CHAPTER 1



01

What a square metre costs

Cost guides price a building to a stated specification and exclude land, fees, VAT and finance. This report follows the same convention unless it says otherwise.

4 of 10

cities in this report with a current, city-specific construction cost from a published guide

Ask what it costs to build a square metre in Lagos or Nairobi and the answer comes from a cost consultant's guide. Ask the same question in Kinshasa, Douala, Libreville, Abidjan, Dakar or Casablanca and the answer is usually an estimate. Of the nine cities in this report, three appear in current international cost guides: Lagos, Nairobi and Dar es Salaam. Five rely on national or regional averages, most of them Rider Levett Bucknall figures for prestige offices. Casablanca has none.

Where the guides overlap, the spread is wide. AECOM's 2024/25 Africa guide prices an average multi-storey residential block at US$2,330 per m^2 in Lagos, US$893 in Dar es Salaam and US$796 in Nairobi. Turner & Townsend's 2026 survey, which averages six building types built to an international specification, puts Lagos at US$1,927 and Nairobi at about US$851, the lowest of the 15 African and Middle East cities it covers.

Cairo, our comparator, sits at the low end on local specification. Knight Frank puts Egyptian apartments at US$720 to 1,270 per m^2 and shell-and-core offices at US$565 to 775. On Turner & Townsend's international basket, Cairo indexes below every Gulf city in the survey.

This chapter sets out the published evidence, assembles it into the Heirstone Build Cost Index 2026 and shows where the specification of a building, rather than its city, explains the spread.

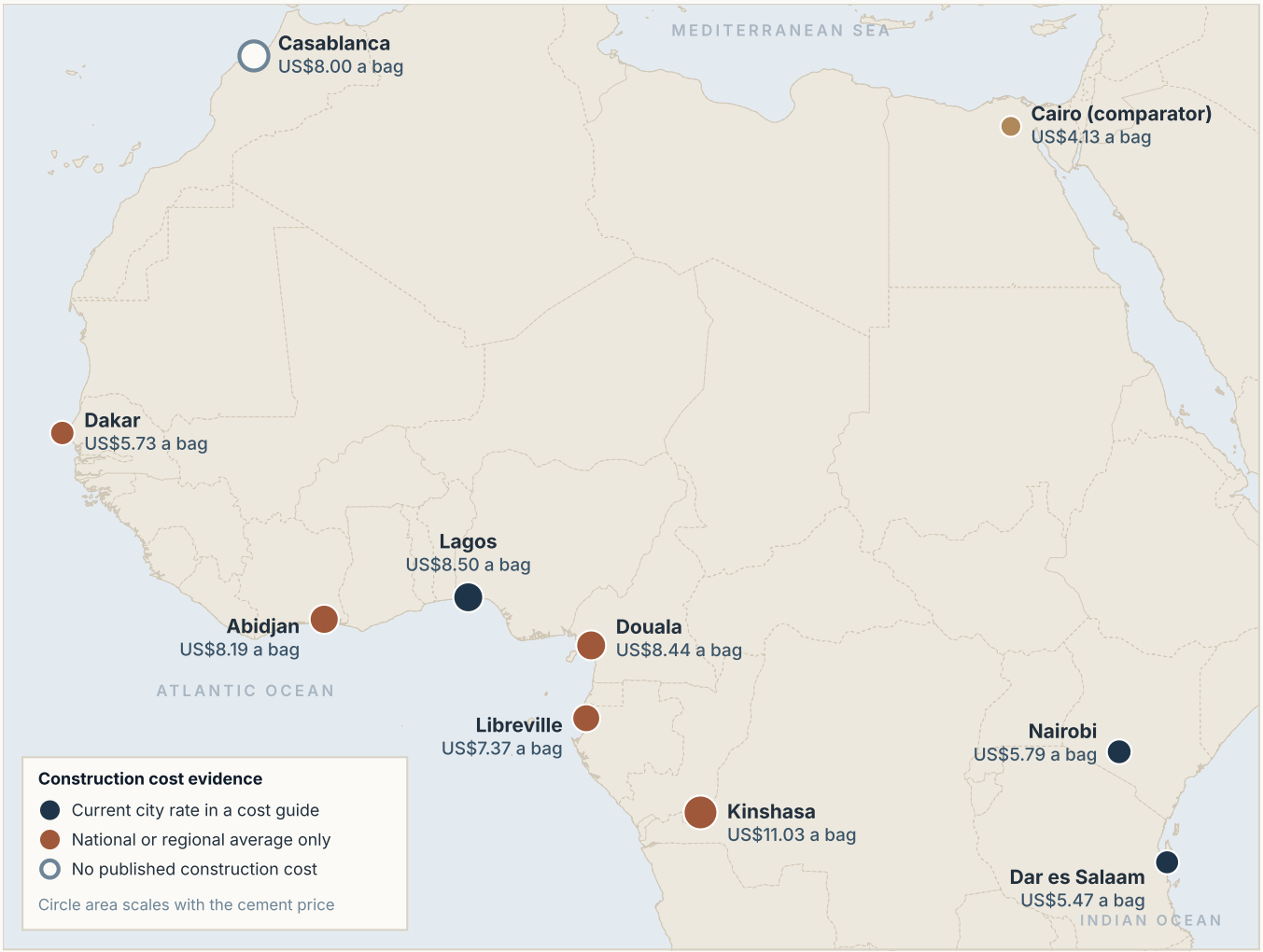
TEN CITIES

Cement is cheapest in Cairo, Dakar and East Africa, and the cost evidence is best in Cairo and East Africa

EXHIBIT 1

A bag of cement costs more than two and a half times as much in Kinshasa as in Cairo

Retail price of a 50 kg bag of cement, US$, latest year published (2023 to 2025), and the type of construction cost evidence available for each city



Source: CAHF, Housing Finance in Africa Yearbook 2024 (country profiles; US$ at CAHF's mid-2024 rates); Arab Finance via Zawya (2026) for Egypt, converted at the World Bank 2025 average rate (EGP49.23). City positions from GeoNames. **Heirstone analysis** for the evidence grades (Exhibit 4).

Two things can be compared across all ten cities: the price of a bag of cement and the quality of the cost evidence. The cheapest bags are in Egypt (US$4.13) and in Dar es Salaam, Dakar and Nairobi (US$5.47 to 5.79); the dearest in Kinshasa (US$11.03), then Lagos, Douala, Abidjan and Casablanca (US$8.00 to 8.50).

Lagos, Nairobi, Dar es Salaam and Cairo are priced city by city in current guides. The five francophone cities south of the Sahara have only national averages reproduced by the Centre for Affordable Housing Finance in Africa (CAHF), mostly Rider Levett Bucknall figures for prestige offices. Morocco publishes no construction cost per m².

US$11.03

Kinshasa, 50 kg bag, 2024 (FC31,627)

US$4.13

Egypt, 50 kg bag, 2025 average (EGP203.5)

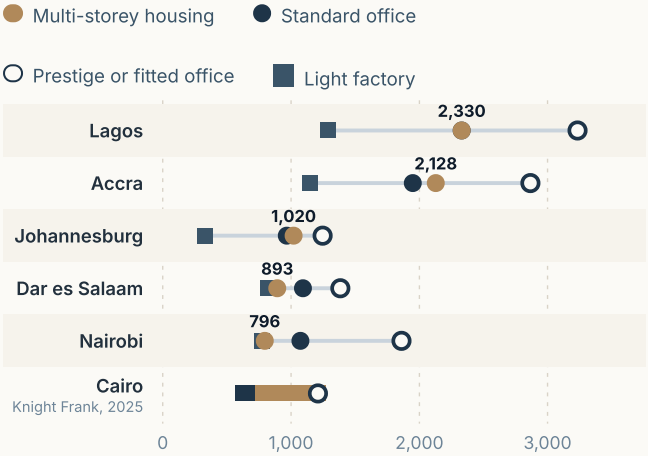
THE PUBLISHED GUIDES

On one guide, Lagos costs almost three times as much as Nairobi. On an international basket, Cairo sits below every Gulf city.

EXHIBIT 2

Lagos and Accra price every building type well above East and Southern Africa; Cairo's local specification sits with Nairobi and Dar es Salaam

US$ per m² by building type, costs at mid-2024 (AECOM) and 2025 (Knight Frank, Cairo ranges)

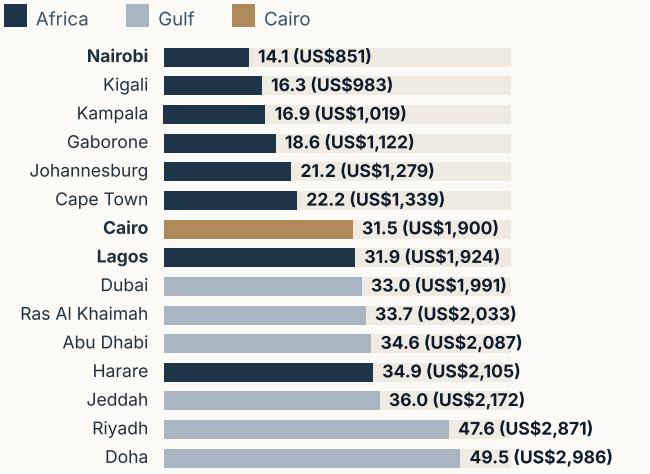


Source: AECOM, Africa Property & Construction Cost Guide 2024/25 (excludes VAT, land, site works, fees and fit-out; 2024 exchange rates); Knight Frank, Egypt construction market release (2025): apartments, shell-and-core offices (range) and fitted offices (up to).

EXHIBIT 3

Nairobi is the cheapest of 15 African and Middle East cities on Turner & Townsend's basket; Cairo is below all six Gulf cities

Location index, London = 100, 2026; in brackets, implied US$ per m² (index x London's US$6,032.5)



Source: Turner & Townsend, Global Construction Market Intelligence 2026 (first-quarter costs and exchange rates, excluding taxes; average of six building types). **Heirstone analysis** for the US$ conversion; Lagos is stated as US$1,927.

The two guides measure different things. AECOM prices local building types in each city. On its 2024/25 figures, Lagos costs 2.9 times Nairobi for a multi-storey residential block and 2.2 times for a standard office, while a light factory in Lagos (US$1,291) costs almost four times one in Johannesburg (US$330). Accra prices close to Lagos. Dar es Salaam and Nairobi sit together, a little below Johannesburg for housing.

Turner & Townsend prices an international basket of offices, hotels, warehouses and apartments in each city and indexes it to London. On that basis, Nairobi is the cheapest market in Africa and the Middle East, at 14.1% of London. Cairo, in the survey for the first time in 2026, indexes at 31.5, just below Lagos and below Dubai (33.0), Abu Dhabi (34.6), Jeddah (36.0), Riyadh (47.6) and Doha (49.5).

Knight Frank's local-specification figures put Cairo lower still. Its range for Egyptian apartments, US$720 to 1,270 per m², brackets AECOM's Nairobi and Dar es Salaam figures and sits 45 to 69% below Lagos.

2.9x

Lagos against Nairobi, multi-storey housing, AECOM 2024/25

45-69%

below Lagos: Egyptian apartments to local specification

THE BENCHMARK

The Heirstone Build Cost Index 2026

Construction cost per m² in US$ from the most recent published source for each city, with the main cost drivers measured in this report. Gold italics are Heirstone estimates for multi-storey housing where no guide rate exists (see page 8). Source codes: AE AECOM 2024/25; TT Turner & Townsend 2024; KF Knight Frank 2025; RL Rider Levett Bucknall prestige-office average, via CAHF (2022, or undated for Douala and Libreville); BC BCEAO national average, via CAHF (building type not stated). n/p not published; n/c not comparable.

EXHIBIT 4

Four of the ten cities have a current guide rate; five rely on national or regional averages; Casablanca has none

US$ per m² unless stated; costs 2022 to 2025; cement 2023 to 2025; interest at the latest published lending rate (2025 or 2026)

CITY	MULTI-STOREY HOUSING	OFFICES: STANDARD / PRESTIGE OR FITTED	LIGHT INDUSTRIAL	CHEAPEST NEW FORMAL HOME, SALE PRICE	CEMENT, US$ PER 50 KG BAG	DUTY + VAT ON IMPORTED CEMENT	INTEREST, 24-MONTH BUILD, % OF HARD COST	GRADE
NORTH AFRICA								
Cairo (comparator)	720-1,270 KF	565-775 / 1,210 KF	n/p	115-138	4.13 (2025)	14%	24.2	A
Casablanca	n/p	n/p	n/p	503	8.00 (2023)	41%	5.0	-
WEST AFRICA								
Lagos	2,330 AE	2,330 / 3,233 AE	1,291	87	8.50 (2023)	29%	20.9-39.3	A
Abidjan	*c.1,080*	1,500 BC	n/p	455	8.19 (2024)	42%	6.8	B
Dakar	*c.1,230*	1,700 RL	n/p	307	5.73 (2024)	42%	5.8	C
CENTRAL AFRICA								
Douala	*c.1,230*	1,700 RL	n/p	402	8.44 (2023)	43%	9.2	C
Libreville	*c.1,280*	1,771 RL	n/p	n/c	7.37 (2024)	42%	23.6	C
Kinshasa	*c.1,300*	1,800 RL	n/p	258	11.03 (2024)	28%	24.9	C
EAST AFRICA								
Nairobi	796 AE	1,073 / 1,861 AE	777	n/p	5.79 (2024)	57%	15.0	A
Dar es Salaam	893 AE	1,093 / 1,385 AE	819	251	5.47 (2023)	59%	15.8	A
REFERENCE CITIES								
Accra	2,128 AE	1,949 / 2,866 AE	1,146	330	5.19 (2024)			A
Johannesburg	1,020 AE	967 / 1,246 AE	330	849	5.64 (2024)			A
Dubai	1,334 TT	1,552 / 1,933 TT	1,035					A
Riyadh	2,132 TT	1,866 / 2,266 TT	1,599					A

Source: Heirstone Build Cost Index 2026, compiled from AECOM (2024/25), Turner & Townsend (2024, 2026), Knight Frank (2025), CAHF Housing Finance in Africa Yearbook 2024 (citing RLB and BCEAO), World Bank WITS tariff data and PwC VAT rates, and central-bank lending rates. Cheapest-home price per m² is **Heirstone analysis** of CAHF figures; Cairo is the SHMFF contractor unit of EGP500,000 in 75 and 90 m² formats. Interest for Lagos is at the prime and maximum lending rates; for Cairo it uses a Heirstone assumption, the central bank's overnight lending rate plus 2 points, after 825 basis points of cuts in 2025 and 2026.

Grade A

City-specific rates from a cost guide published in 2024 to 2026 (AECOM, Turner & Townsend, Knight Frank).

Grades B and C

A central-bank national average for 2024 that states no building type (B), or a Rider Levett Bucknall regional or national average for prestige offices (C).

Heirstone estimate

The published average multiplied by 0.72, the median ratio of housing to prestige-office cost in AECOM's five African cities. Indicative only.

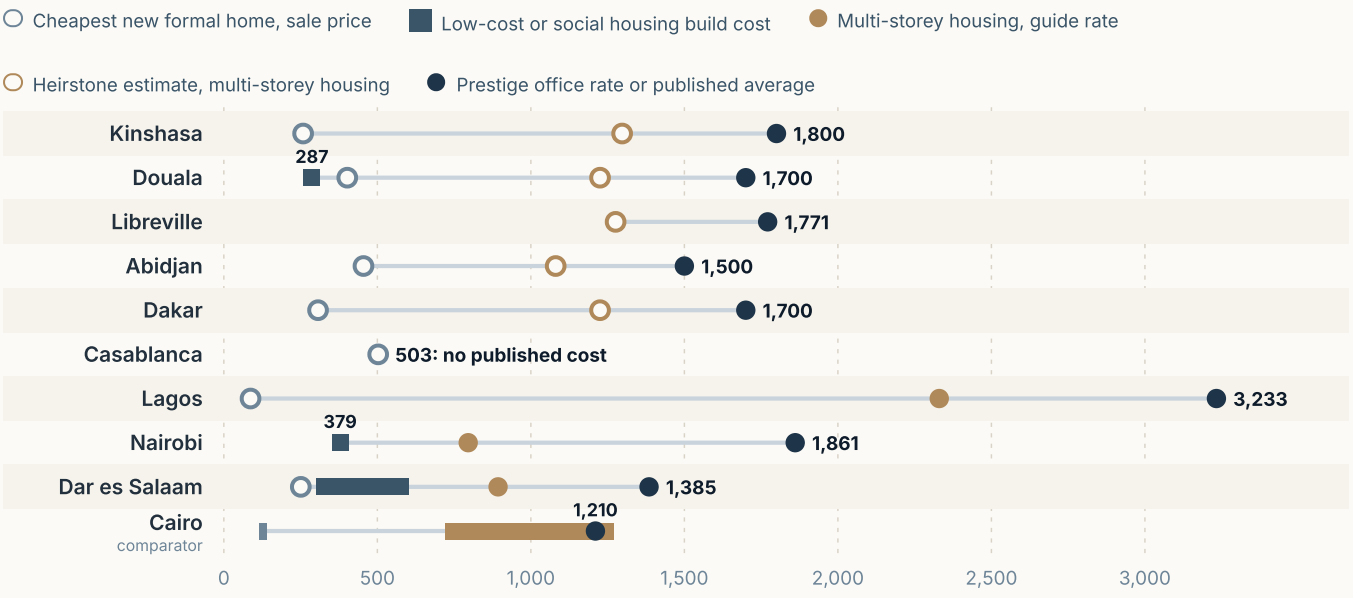
READING THE INDEX

Specification moves the price of a square metre further than geography does

EXHIBIT 5

The averages published for Central and West Africa, mostly prestige-office figures, are three to seven times what the cheapest formal homes sell for

US$ per m²: sale price of the cheapest newly built formal home, low-cost build costs, multi-storey housing and published averages; 2021 to 2025



Source: CAHF Housing Finance in Africa Yearbook 2024 (US$ at mid-2024 rates; cheapest homes 2021 to 2024; Douala state social housing; Kenya and Tanzania national averages); AECOM 2024/25; RLB (2022 or undated) and BCEAO via CAHF; Knight Frank (2025). Sale price per m² and estimates are **Heirstone analysis**. Libreville's cheapest-home figures conflict in the source and are not shown.

In Douala, the state builds social housing for about CFA175,000 per m² (US$287). The national average construction cost that CAHF reproduces for Cameroon is CFA1,037,051 per m² (US$1,700). This gap within one city, a factor of 5.9, exceeds the gap between any two African cities in the guides. The pattern holds wherever the data allow: the cheapest newly built formal homes sell for US$251 to 503 per m² in Dar es Salaam, Kinshasa, Dakar, Douala, Abidjan and Casablanca, far below the published averages.

CAHF attributes the averages for Kinshasa, Douala, Dakar and Libreville to Rider Levett Bucknall, whose 2022 Africa survey prices prestige high-rise offices (US$1,800 per m² on average in Central Africa); Abidjan's, from the BCEAO, states no building type. Used as the cost of housing, they overstate it. Applying the median ratio of housing to prestige-office cost in AECOM's African cities, 0.72, gives indicative costs of about US$1,080 to 1,300 per m² for multi-storey housing in the five cities; the full range of ratios widens that to US$640 to 1,470.

Egypt shows what volume does to price. The Social Housing and Mortgage Finance Fund's cheapest contractor-built unit, EGP500,000, works out at about US$115 to 138 per m² of sale price for its 75 and 90 m² units, and the fund had completed 685,000 fully finished units by mid-2024.

5.9x

Douala: the published average against state social housing, per m²

685,000

fully finished units completed by Egypt's Social Housing and Mortgage Finance Fund by mid-2024

CHAPTER 2

02

Cement is the smaller part of the bill

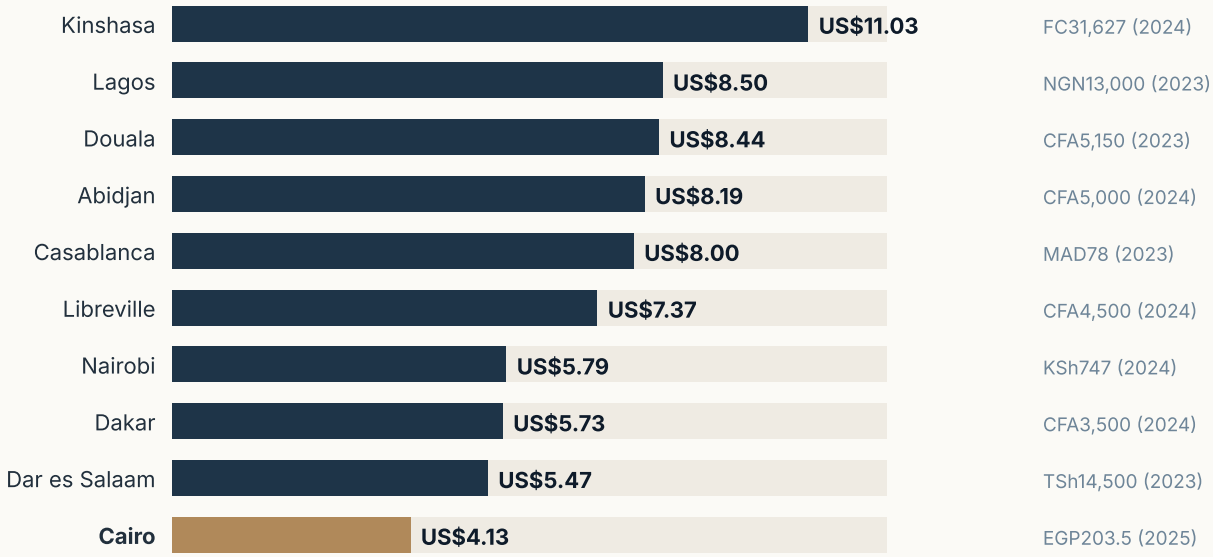
Cement prices are the most quoted cost in African construction. They vary widely: a 50 kg bag costs more than two and a half times as much in Kinshasa as in Cairo. Yet cement is a modest share of what a building costs, and spare kiln capacity has not, on its own, brought prices down.

This chapter measures the cement and steel in a square metre, compares their prices across the ten cities and sets out why Egypt, a large producer that runs its plants hard, is the regional price benchmark.

EXHIBIT 6

A bag of cement in Egypt costs about half what it does in Lagos, Douala, Abidjan or Casablanca

Retail price of a 50 kg bag, US$ at the source's rate, with the local price and year



Source: CAHF Housing Finance in Africa Yearbook 2024 (US$ at CAHF's mid-2024 rates; local prices from Mustard Insights, The Sun, Cemnet, Jiji, Dangote Cement and HZECH as cited by CAHF); Egypt: Arab Finance via Zawya (2026), 2025 average of EGP4,070 a tonne, converted at EGP49.23 (World Bank, 2025 average). **Heirstone analysis** for the Egypt per-bag figure.

Later prices confirm the order. Regulated prices in Douala were cut to CFA5,000 a bag for 42.5R cement at the end of 2023, and bags sold for CFA5,100 to 5,300 (about US$9.12 to 9.48) in 2025. Côte d'Ivoire cut its ex-factory ceilings by CFA7,000 a tonne from the start of 2025, to CFA78,000 for 42.5 grade (about US$129). In Nigeria, the competition regulator recorded bags at NGN9,300 to 9,700 early in 2026 and NGN13,000 to 15,000 in some areas by mid-year.

CAHF's Egypt profile lists EGP2,050 for a bag of cement, a figure in line with Egyptian prices per tonne at the time. We use the 2025 market average instead: EGP4,070 a tonne, or EGP203.5 a bag, about US$4.13 at the 2025 average exchange rate. At the 2026 market rate it is about US$3.85. Gabon's profile gives two prices, CFA4,500 and CFA5,500; we show the lower, from its key figures.

In local currency the moves are larger. CAHF records Nigerian cement rising from NGN2,400 to NGN14,000 a bag between 2019 and 2024, almost six times in naira. In Kenya, cement and lime prices fell 1.39% in the third quarter of 2025 while steel and reinforcing bars rose 5.2%.

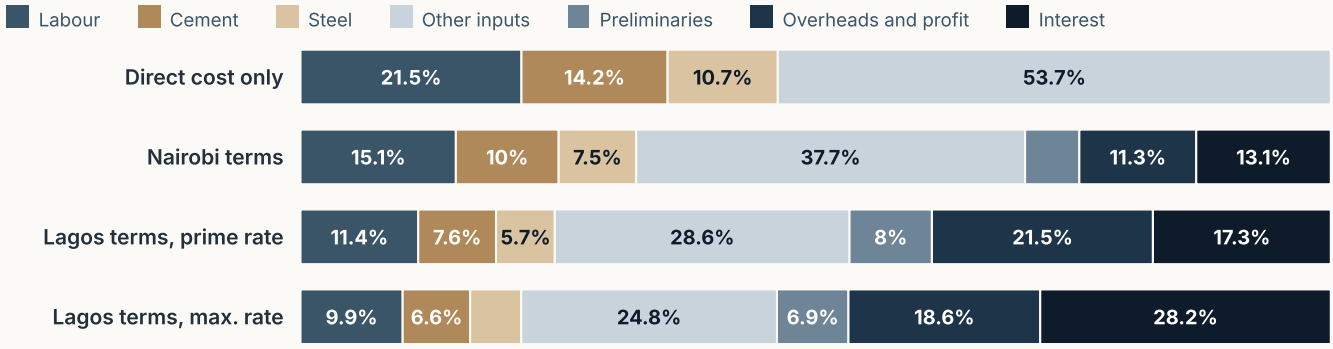
WHAT IS IN A SQUARE METRE

Once overheads and interest are added, cement is a tenth of the cost of a square metre in Nairobi and less than that in Lagos

EXHIBIT 7

The same building priced on Lagos terms carries 47 to 54% of its cost in preliminaries, overheads, profit and interest, against 30% on Nairobi terms

Share of all-in cost: direct inputs at Kenya's building cost index weights, plus each city's preliminaries and contractor overheads and profit (2,500 m² job) and interest on a 24-month, fully debt-funded build



Source: KNBS, Construction Input Price Indices (index weights, base 2019); Turner & Townsend, Global Construction Market Intelligence 2026 (preliminaries and overheads and profit); CBK and CBN lending rates (2026). **Heirstone analysis.** The direct cost mix is Kenya's for both cities; the exhibit isolates the effect of markups and interest.

Kenya's statistics office is the only one in the sample that publishes the weights behind its building cost index. Labour carries 21.45%, cement and lime 14.18% and steel and reinforcing bars 10.69%. The rest is aggregates, blocks, timber, finishes, services and plant. On those weights, a 50% rise in the price of cement raises the direct cost of a building by about 7%. The gap between Kinshasa's bag and Nairobi's, a factor of 1.9, would add about 13%.

Direct cost is only the start. A contractor adds preliminaries, the cost of running the site, and then overheads and profit. A lender adds interest while the building is going up. On Turner & Townsend's rates for a 2,500 m² job, a Nairobi contractor adds 7.5% for preliminaries and 15% for overheads and profit; a Lagos contractor adds 15% and 35%. At Kenya's average lending rate of 14.34%, a two-year build carries interest of 15% of hard cost; at Nigeria's prime and maximum rates, 21% and 39%.

Put together, these layers make the same building 32 to 52% dearer on Lagos terms than on Nairobi terms before any difference in materials, labour or design. On Lagos terms at the prime rate, cement falls to 7.6% of the all-in cost, while preliminaries, overheads, profit and interest make up 47%.

14.18%

weight of cement and lime in Kenya's building cost index

32-52%

added by Lagos markups and interest to the same direct cost, against Nairobi terms

> *On Lagos terms, a 20% cut in the price of cement lowers the all-in cost of a square metre by about 1.5%. Borrowing at the prime rate rather than the maximum lowers it by 13%.*

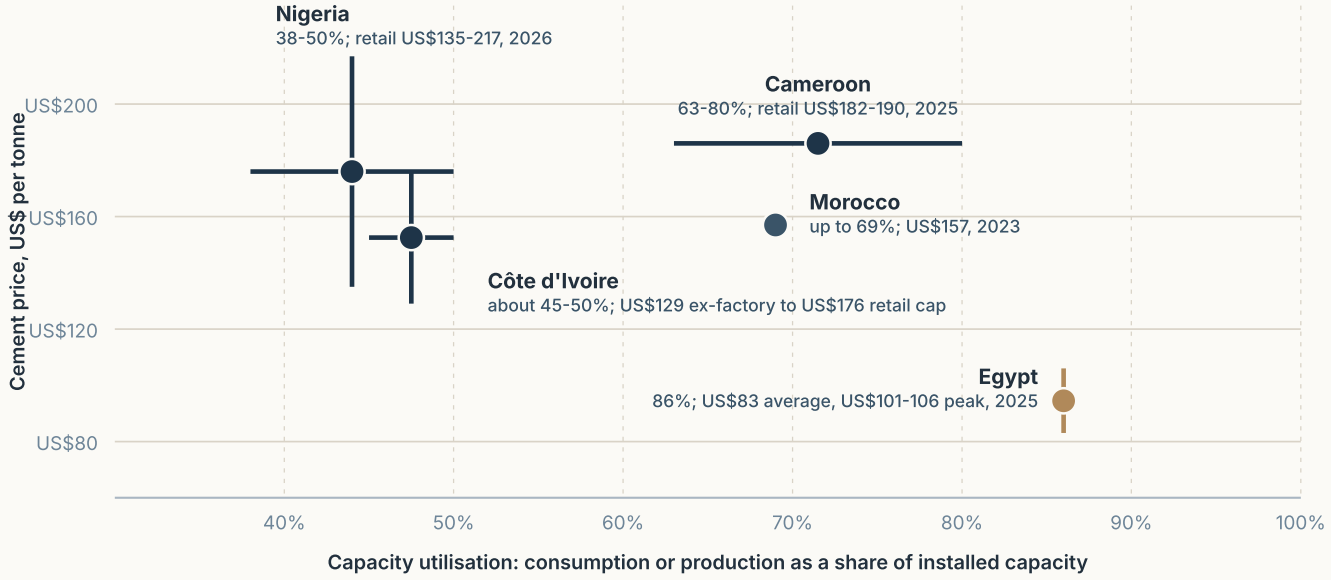
CAPACITY AND PRICE

Spare kilns have not brought cheap cement. Egypt runs its plants harder and sells for less.

EXHIBIT 8

The markets with the most idle capacity do not have the lowest prices

Cement capacity utilisation and price per tonne, latest available; lines show published ranges



Source: FCCPC via CemNet (2026); Dangote Cement FY2025 release; Federation of Egyptian Industries via Global Cement (2025); CemNet (2026); Arab Finance via Zawya (2026); Fraternité Matin (2025); Afrique-sur7 (2025) for the Abidjan retail cap; Daba Finance (2025) and Erooamba (2026), business websites, for Côte d'Ivoire capacity and use; Business in Cameroon and CemNet (2025); Holcim Maroc and Heidelberg Materials (2025); APC via Le Matin (2026); CAHF (2024). US$ at US Treasury or World Bank rates for the year. **Heirstone analysis.** Morocco covers the two largest producers only, so its utilisation is an upper bound.

Nigeria's competition commission estimates national capacity at 60 to 65 Mt a year against consumption of 25 to 30 Mt, a utilisation of 38 to 50%, and in 2026 it opened an investigation into cement pricing, noting that the surplus had not produced the fall in prices it would expect in a competitive market. Producers point to their costs. Nigerian manufacturers spent NGN1.34trn on their own power in 2025, while average grid supply fell from 16.7 hours a day in the first half of the year to 13.1 in the second, and Dangote Cement says it could not find a vessel for a 1,000 t shipment to Ghana.

Côte d'Ivoire has about 19 to 20 Mt of capacity for 7 to 10 Mt of consumption, according to two business websites, Daba Finance (2025) and Erooamba (2026), and Cameroon 10 to 12.7 Mt for about 8 Mt. Both rely largely on imported clinker; Cameroon bought 2.41 Mt in 2023.

Egypt sits at the other corner of the chart. It has 76 Mt of capacity on 46 lines and produced about 65 Mt in 2025, 18% more than in 2024, as domestic consumption rose 13.4% to 54 Mt. The average price for the year was EGP4,070 a tonne, about US$83; even at the mid-year peak it was about US$101 to 106. Exports of cement and clinker reached 18.5 Mt, with Côte d'Ivoire and Ghana among the named destinations, and Titan Egypt is adding export silos in Alexandria to reach 1.3 Mt of export capacity by 2028.

86%

Egyptian cement capacity in use in 2025 (65 Mt of 76 Mt)

38-50%

Nigeria: consumption against installed capacity, regulator estimate

18.5 Mt

Egyptian cement and clinker exports, 2025

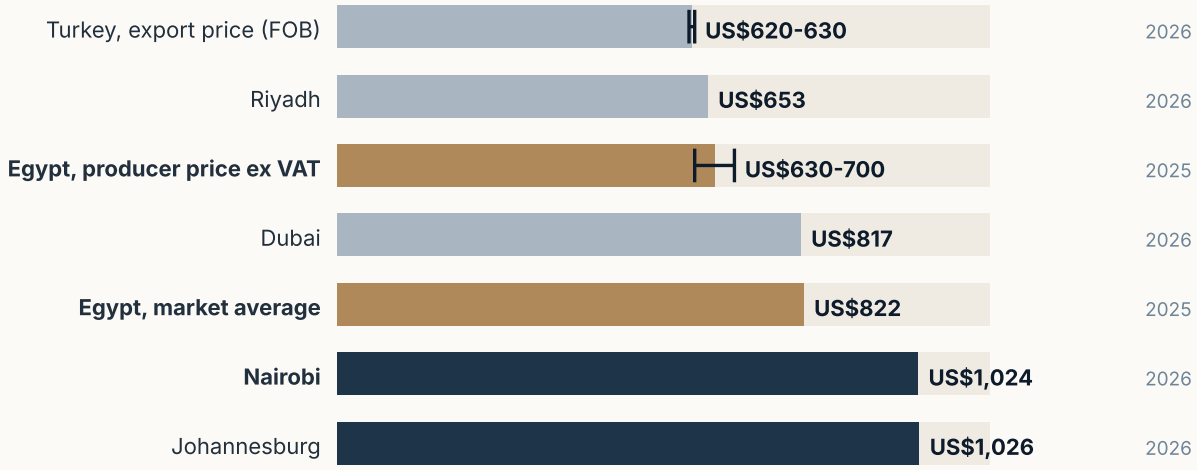
STEEL

Egyptian rebar trades close to world export prices and at least a fifth below Nairobi and Johannesburg

EXHIBIT 9

A tonne of reinforcing bar costs about US$1,025 in Nairobi and Johannesburg, against US$630 to 822 in Egypt

Reinforcing bar, US$ per tonne: contractor rates for a 120 t job (Turner & Townsend), Egyptian producer and market prices, and the Turkish export reference



Source: Turner & Townsend, GCMI 2026 cost data (16 mm bar, 120 t job, first-quarter 2026 rates, excluding taxes); Arab Iron and Steel Union producer price lists (2025, ex VAT at US Treasury rates; Heirstone analysis) and weekly prices (2026); Arab Finance via Zawya (2026), 2025 average of EGP40,482 at EGP49.23; the source does not state whether this includes 14% VAT (without it, about US$721). **Heirstone analysis.**

Steel is the second structural input, about 11% of direct cost on Kenya's weights, and here Egypt's advantage is scale. It produced 10.6 Mt of crude steel in 2025, 19th in the world and about 36% of Africa's output of 29.8 Mt, all of it by the electric route. Egypt used 10.0 Mt of finished steel in 2025, more than twice South Africa's 4.4 Mt, or 84.3 kg a head against 29.2 kg for Africa as a whole.

Egypt's 2025 average rebar price, EGP40,482 a tonne, was about US$822, level with Turner & Townsend's Dubai rate (US$817) and about 20% below its Nairobi (US$1,024) and Johannesburg (US$1,026) rates, which exclude taxes. Arab Finance does not say whether its average includes VAT; if it does, as Egyptian producer price lists do, the gap is about 30%. Producer list prices fell late in 2025: Ezz Steel's price excluding 14% VAT was about EGP30,000 a tonne, roughly US$630, close to the Turkish export reference of US$620 to 630.

In East Africa, steel is also a border question. The East African Community charges 35% duty on rebar, and Kenya adds a 17.5% levy on imported semi-finished steel inputs; in 2025 its parliament rejected a proposal to cut that levy to 10%. Ready-mixed 30 MPa concrete costs US$103 per m^3 in Nairobi and US$158 in Johannesburg, against US$87 in Dubai and US$80 in Riyadh.

36%

Egypt's share of Africa's crude steel output in 2025 (10.6 of 29.8 Mt)

84.3 kg

finished steel used per Egyptian in 2025; Africa 29.2 kg

20-30%

Egypt's 2025 average rebar price below Nairobi's contractor rate, with or without VAT

> *On Kenya's weights, a 20% cut in the price of steel lowers the direct cost of a building by about 2%.*

CHAPTER 3

03 Duty, levy and haulage

Imported cement pays duty and VAT before it reaches a site, and every bag, imported or not, pays for the road. In East Africa the border adds the most; in Gabon and the DR Congo, the road.

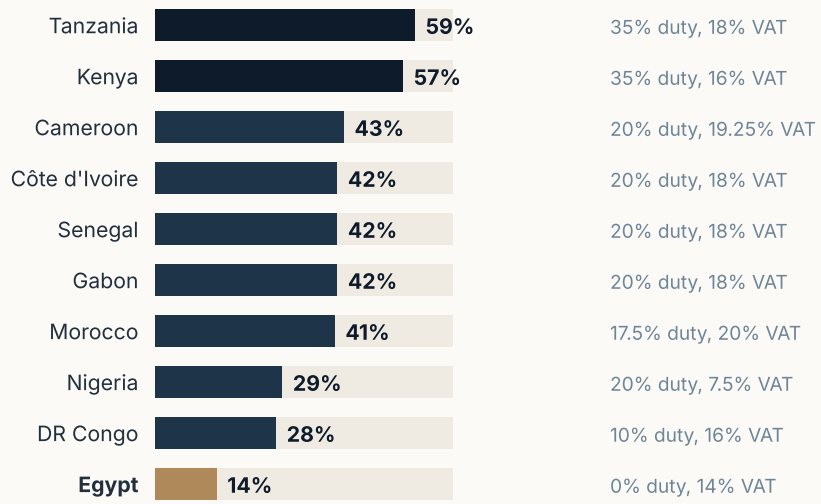
The East African Community adopted 35% as a fourth band of its common external tariff in 2022 and applies it to Portland cement, rebar and sanitary ware. With VAT on top, a tonne of imported cement costs 57% more than its landed value in Kenya and 59% more in Tanzania. The ECOWAS and CEMAC customs unions charge 20%, which with VAT adds 41 to 43% in the CFA franc countries; Nigeria's lower VAT keeps its wedge at 29%. On its latest published schedule, Egypt charges no duty on cement, clinker or rebar, so the only addition is its 14% VAT.

VAT also falls on cement made at home, so duty is the part the border alone adds: 35% in the EAC, 20% in ECOWAS and CEMAC, 17.5% in Morocco, 10% in the DR Congo and nothing in Egypt.

EXHIBIT 10

Duty and VAT add 57 to 59% to imported cement in East Africa and 14% in Egypt

Duty plus VAT as a share of the landed (CIF) value of imported Portland cement; other levies excluded



Source: World Bank WITS and UNCTAD TRAINS, applied MFN tariffs (2023; Egypt 2019 and DR Congo 2020, the latest available); PwC Worldwide Tax Summaries, standard VAT rates (2026). **Heirstone analysis:** CIF x (1 + duty) x (1 + VAT).

AT THE BORDER

35%
the fourth band of the East African Community's common external tariff, adopted in 2022 and applied to cement and rebar

0%
Egypt's duty on cement, clinker and rebar in its latest published schedule

17.5%
Kenya's levy on imported clinker, after which clinker imports fell 93% in a year

+60%
cement in Gabon's interior against the Libreville area, on the official price list

41-43%
duty and VAT on imported cement in the CFA franc countries

TARIFFS AND CLINKER

The next saving lies in clinker, and four countries are already moving to make their own

EXHIBIT 11

Clinker enters most markets at 10%; finished cement and rebar pay 20% in West and Central Africa and 35% in East Africa

Applied import duty and standard VAT, % of value, latest year published

CUSTOMS UNION	COUNTRY	CLINKER	CEMENT	REBAR	VAT	DUTY + VAT ON CEMENT	OTHER MEASURES
ECOWAS	**Nigeria**	10	20	20	7.5	29	Surplus capacity; clinker shipped to grinding plants in Cameroon and Ghana
ECOWAS (WAEMU)	**Côte d'Ivoire**	10	20	20	18	42	Ex-factory price ceilings, cut from 2025
ECOWAS (WAEMU)	**Senegal**	10	20	20	18	42	New kiln replaces 300-400 kt of clinker imports
CEMAC	**Cameroon**	10	20	20	19.25	43	New CEEAC-CEMAC tariff from 2026: cement in the 5-20% band
CEMAC	**Gabon**	10	20	20	18	42	Official price list; clinker import ban planned from 2027
National	**DR Congo**	5	10	20	16	28	
EAC	**Kenya**	10	35	35	16	57	17.5% levy on clinker and semi-finished steel
EAC	**Tanzania**	10	35	35	18	59	
National	**Morocco**	2.5-17.5	17.5	2.5-40	20	41	
National	**Egypt**	0	0	0	14	14	Exports capped at 30% of production

Source: World Bank WITS and UNCTAD TRAINS (MFN applied tariffs, 2023; Egypt 2019; DR Congo 2020); EAC (2022); PwC (VAT, 2026); Business Daily (2025); CemNet (2025, 2026); Business in Cameroon (2026); Global Cement (2025); Fraternité Matin (2025); Vicat (2026); CAHF (2024). **Heirstone analysis** for the duty and VAT wedge.

Cement is made by grinding clinker, the product of the kiln, and clinker crosses borders at a lower duty than cement: 10% against 20 to 35%. That is why many plants in the sample are grinding units near a port, and why the next saving lies upstream, in making clinker at home. Four countries are moving in that direction.

Kenya introduced a 17.5% levy on imported clinker, on top of the 10% common tariff. Clinker imports fell 93% in a year, from 148,019 t in 2023 to 10,337 t in 2024, and Bamburi has signed a US$250m contract to add 1.6 Mt of clinker capacity at Kwale. In Senegal, Vicat's new kiln 6 at Sococim reached nominal capacity early in 2026, replacing 300 to 400 kt of imported clinker a year at a saving of about €20 a tonne. Cameroon, which imported 2.41 Mt of clinker in 2023 worth CFA87.7bn (US$156.1m), opened its first clinker plant at Figuil in 2025, and a plant at Kribi makes a low-clinker cement. Gabon plans to ban clinker imports from 2027, as CIMAF plans an investment of over US$45m at Owendo that would take capacity to about 1.85 Mt against national demand of about 0.9 Mt.

Finishes pay more at the border than structure does. Ceramic tiles and sanitary ware carry 20% in ECOWAS and 30% in CEMAC; the EAC charges 35% on sanitary ware and Morocco 40% on both.

-93%

Kenya's clinker imports in the year after its 17.5% levy

€20/t

saving on each tonne of clinker that Senegal's new kiln makes instead of importing

2.41 Mt

clinker imported by Cameroon in 2023, at about US$65 a tonne

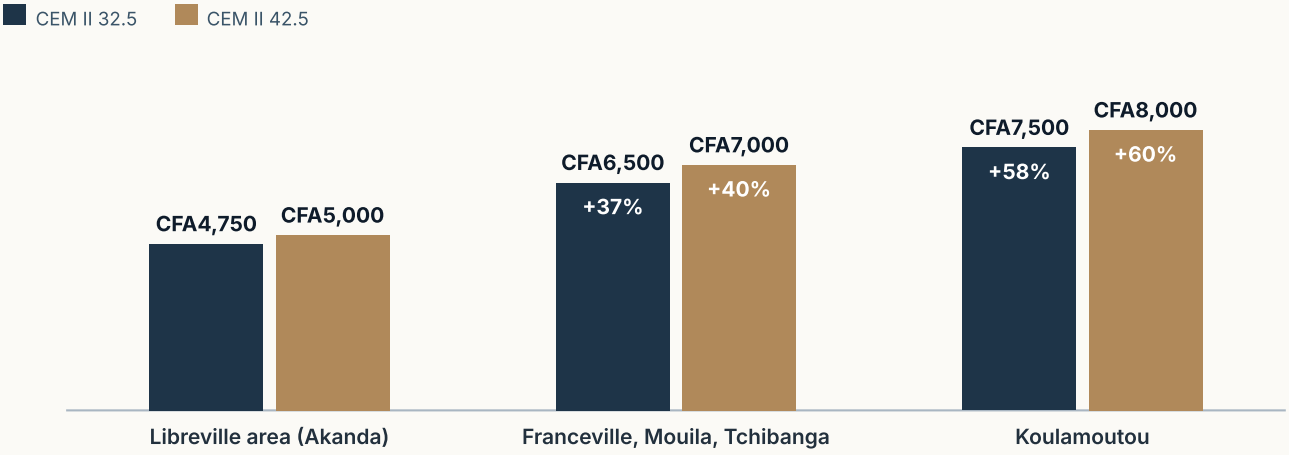
THE ROAD

Distance can cost more than duty: in Gabon's interior, a bag costs up to 60% more than around Libreville

EXHIBIT 12

Gabon's official price list adds 37 to 60% to a bag of cement between the Libreville area and the interior

Regulated retail price of a 50 kg bag by town and grade, CFA francs; percentages against the Libreville area



Source: Government of Gabon, cement price order (2023), via CemNet and L'Union (2023). Franceville, Mouila and Tchibanga: lower end of CFA6,500 to 6,800 for CEM II 32.5. **Heirstone analysis** for the premiums.

Gabon publishes what few governments do: an official cement price for each town. Around Libreville, CEM II 32.5 cement retails at CFA4,750 a bag. In Franceville, Mouila and Tchibanga it is CFA6,500 to 6,800 and in Koulamoutou CFA7,500. The difference, 37 to 60%, is comparable to the whole duty and VAT wedge on imported cement in the CFA franc zones, and it is paid on cement ground in the country.

Kinshasa shows the same effect over time. Much of its cement comes by road from plants in Kongo Central, 160 to 190 km away in a straight line. In 2025, after a transport strike and government intervention, the price of a bag fell 52%, from US$21 to US$11. During an earlier strike, in 2024, a bag of grey cement rose from its usual FC45,000 to FC55,000, according to a local report.

Côte d'Ivoire writes distance into its regulated price: CFA102,000 to 106,250 a tonne in Abidjan, plus CFA30 per tonne-kilometre elsewhere. A site 300 km inland pays about 9% more, or CFA450 a bag. In Tanzania, the prime minister has said that western regions such as Kigoma depend on cement shipped long distances from Mtwara and Tanga, and that a plant in Kigoma has reached implementation.

We could not find published port handling or inland haulage costs per tonne of building material for any of the ten countries. For an investor choosing between sites, the distance to the nearest kiln or grinding plant belongs in the feasibility study alongside the price of land.

-52%

Kinshasa cement price after a 2025 transport strike ended: US$21 to US$11 a bag

CFA30

per tonne-kilometre: Côte d'Ivoire's regulated surcharge outside Abidjan

160-190 km

from Kinshasa to the Kongo Central cement plants, straight line (Heirstone analysis)

CHAPTER 4

04 The price of time

A building costs money for as long as it is unfinished. In five of the nine cities, that cost is larger than all the cement in the building.

Interest during construction rarely appears in cost guides, which price the building and leave the finance to the developer. Yet the rate at which a developer borrows, and the months for which it borrows, decide a large part of what a finished square metre costs. Across the ten cities, the average lending rate runs from about 5% in Casablanca and Dakar to 33% for Nigerian borrowers paying the maximum rate.

Margins work the same way. A contractor in a market where credit is short and payment slow prices that risk into its overheads and profit. Turner & Townsend's figures for a small job show Lagos contractors adding 35% against 15% in Nairobi.

Currency adds a third layer for anyone who counts in dollars. Turner & Townsend's Lagos average rose 63% in dollars between its 2025 and 2026 surveys while costs in naira rose 23%; the naira's own recovery explains only part of the gap.

This chapter measures each layer and shows how much time is worth in each city. That figure decides where faster building methods pay.

* Cairo: Heirstone assumption of the central bank's overnight lending rate plus 2 points, the margin CAHF reports for Egyptian mortgages, after 825 basis points of cuts in 2025 and 2026.

5.0%
CASABLANCA
interest on a 24-month build, as a share of hard cost, at the average real-estate lending rate

20.9-39.3%
LAGOS
the same build at Nigeria's average prime and maximum lending rates

4.50%
BEAC POLICY RATE
shared by Cameroon and Gabon, where bank credit cost 9.03% and 21.51%

3.3 years
KENYA
average time to build a house in 2023, which lifts interest from 15% to 24% of hard cost

AVERAGE LENDING RATES USED IN THIS CHAPTER, %, LATEST PUBLISHED

5.06	5.86	6.78	9.03	14.34	15.05	19.29	21.51	22.0	22.58	33.16
Casablanca	Dakar	Abidjan	Douala	Nairobi	Dar es Salaam	Lagos, prime	Libreville	Cairo*	Kinshasa	Lagos, max.

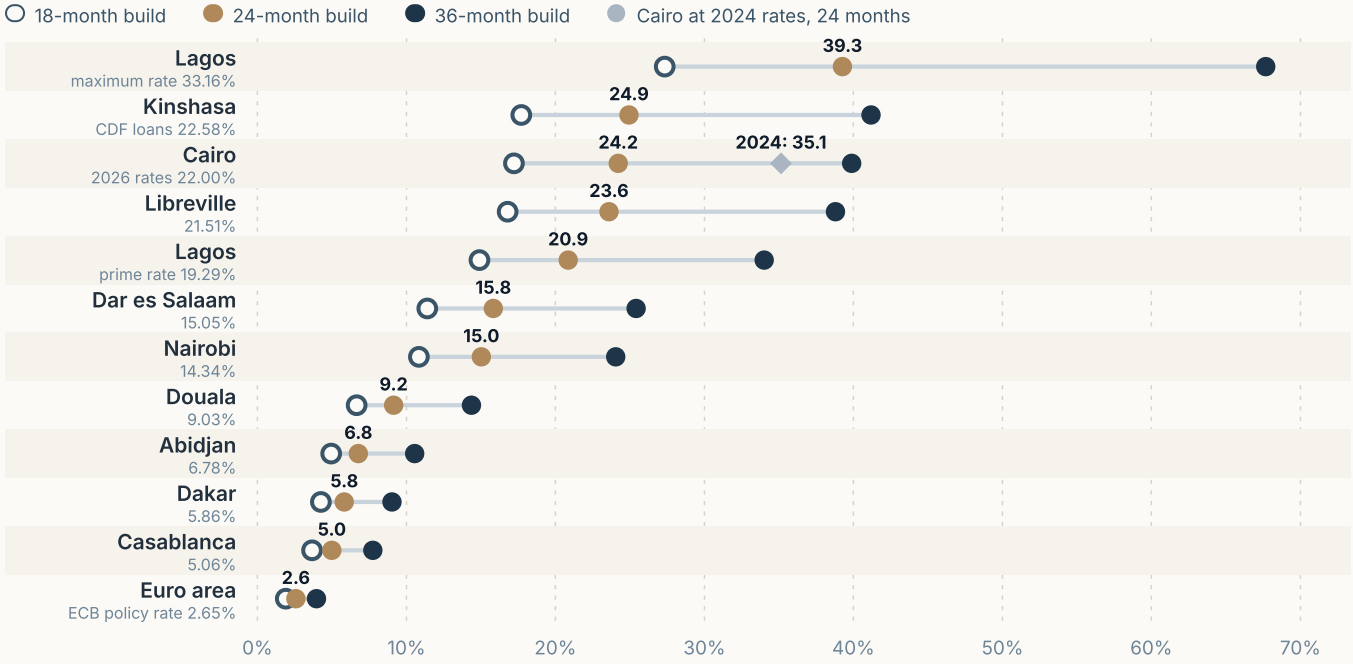
INTEREST DURING CONSTRUCTION

Interest on a two-year build runs from 5% of hard cost in Casablanca to 39% in Lagos

EXHIBIT 13

In five of the nine cities, interest on a two-year build exceeds the weight of all the cement in the building

Interest during construction as % of hard cost, at the latest published average lending rate, for builds of 18, 24 and 36 months, fully debt-funded



Source: BCC (2026); BEAC (first quarter 2026); CBN via ThisDay (2026); BCEAO and CAHF (2024, 2025); CBK (2026); Bank of Tanzania (2026); Bank Al-Maghrib via Challenge.ma (2026); CBE via Amwal Al Ghad and FocusEconomics (2025, 2026) and CAHF (2024); ECB (2026). **Heirstone analysis:** equal monthly drawdowns, monthly compounding, interest rolled up to completion. Cairo uses a Heirstone assumption: the central bank's overnight lending rate plus 2 points, the mortgage margin CAHF reports.

At each city's latest average lending rate, a fully debt-funded 24-month build carries interest of 5.0% of hard cost in Casablanca, 5.8% in Dakar and 6.8% in Abidjan, 9.2% in Douala, 15 to 16% in Nairobi and Dar es Salaam, 24 to 25% in Libreville and Kinshasa, and 20.9 to 39.3% in Lagos. Cement and lime, at 14.18% of Kenya's index, weigh less than the interest in Kinshasa, Lagos, Libreville, Dar es Salaam and Nairobi.

Time multiplies the rate. CAHF reports that building a house in Kenya took an average of 3.3 years in 2023. Stretching a programme from two years to three lifts Kenyan interest from 15.0% to 24.0% of hard cost. In Kinshasa it adds 16 points, though 95% of Congolese bank loans are in dollars, so the franc rate is an upper bound for most developers.

Egypt shows how fast the layer can shrink. The Central Bank of Egypt cut its key rates by 725 basis points in 2025 and by a further 100 points early in 2026, as inflation eased to 12.2% by mid-2026. On our assumption of the lending rate plus 2 points, interest on a 24-month Cairo build fell from 35.1% to 24.2% of hard cost, and each further 100 basis points of easing removes about 1.2 points more.

24.0%

interest on a three-year Nairobi build, against 15.0% over two years

-10.9 pts

interest on a two-year Cairo build since the central bank began cutting in 2025

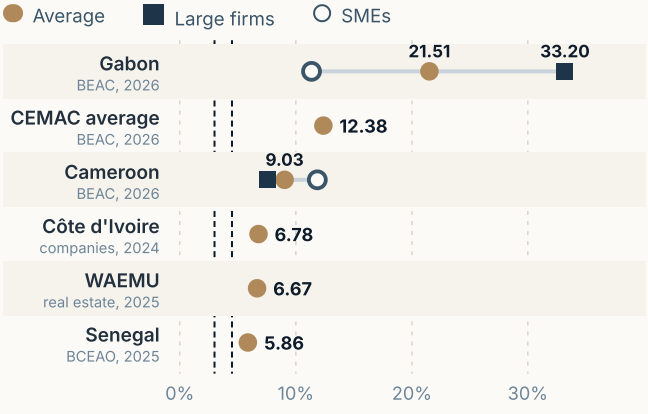
CREDIT AND MARGIN

Inside one currency union, the price of credit varies more than the price of cement

EXHIBIT 14

Cameroon and Gabon share one policy rate; their bank credit costs 9.03% and 21.51%

Average lending rates by borrower, %, in the two CFA franc unions; dashed lines are the policy rates (BEAC 4.50%, BCEAO 3.00%)

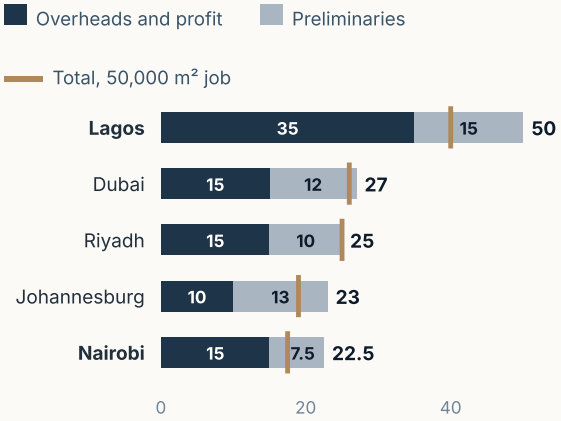


Source: BEAC, report on lending rates, first quarter 2026; BCEAO, annual report on banking conditions in WAEMU (2025) and data cited by CAHF (2024); BCEAO via the Journal de l'Economie Sénégalaise (2025).

EXHIBIT 15

Lagos contractors add 50 points to a small job; Nairobi's add 22.5

Contractor overheads and profit plus preliminaries, % of construction cost, 2,500 m² job; gold mark shows the total on a 50,000 m² job



Source: Turner & Townsend, Global Construction Market Intelligence 2026, construction market metrics.

Cameroon and Gabon share a central bank, a currency and, since mid-2026, a 4.50% policy rate. In the first quarter of 2026, Cameroon's credit institutions lent at an average of 9.03%. Gabon's banks charged 21.51%, up from 14.48% a year earlier, and large Gabonese firms paid 33.20%, twice the rate of a year before. The BEAC's own report names Gabon as the most expensive country in the union for bank credit. Long money is scarce in both: long-term loans were 4.71% of new credit in Cameroon and 1.85% in Gabon, so most construction is funded with short lines that have to be rolled over.

The West African union is more even. Average bank lending was 6.69% in 2025, real-estate loans 6.67%, and Senegal's average 5.86%. With WAEMU inflation at 1.3% in 2026, those are real rates of about 4.5 to 5.5%, against 7.5% in Kenya. Within the CFA franc zone, cheap credit is a West African feature: Gabon's average rate is more than three times Senegal's.

Margins add a second layer. On Turner & Townsend's figures, a Lagos contractor adds 35% for overheads and profit and 15% for preliminaries on a 2,500 m² job, and still 30% and 10% on a 50,000 m² job. Nairobi's figures are 15% and 7.5%, below Dubai and Riyadh. Bigger packages lower the markup in Lagos, Nairobi and Johannesburg, and in Lagos the saving from scale alone is about 8% of the contract sum.

1.85%

long-term loans as a share of new bank credit in Gabon, first quarter 2026

-8%

Lagos contract sum on a large package against a small one, from overheads, profit and preliminaries

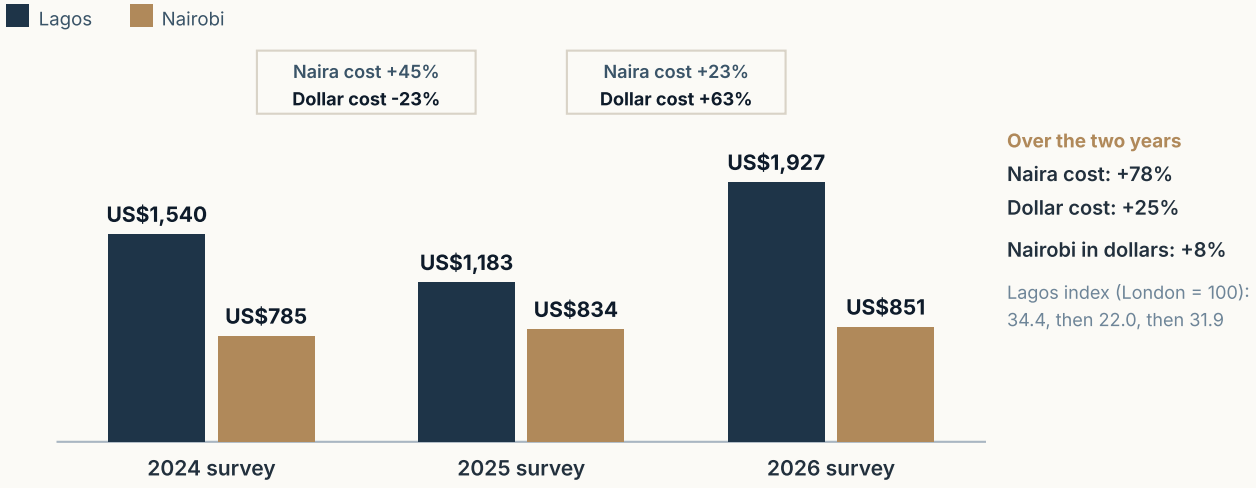
CURRENCY AND LABOUR

In dollars, Lagos fell by almost a quarter in one survey year and rose by almost two-thirds in the next, while naira costs rose in both

EXHIBIT 16

Lagos's dollar cost per m² has swung by more than the naira explains; Nairobi's barely moves

Turner & Townsend average construction cost, US$ per m², by survey year, with Lagos cost escalation in naira



Source: Turner & Townsend, ICMS 2024 (via Kenyans.co.ke), GCMI 2025 (via Creamer Media and Turner & Townsend) and GCMI 2026; market metrics sheets for escalation and the location index. Each survey converts local costs at that year's first-quarter exchange rates; naira per US$ at the end of the first quarter was 1,387 (2024), 1,530 (2025) and 1,375 (2026), US Treasury. **Heirstone analysis** for the changes.

Turner & Townsend's Lagos average fell from US$1,540 per m² in its 2024 survey to US$1,183 in 2025, then rose to US$1,927 in 2026, while it recorded cost escalation of 45% and 23% in naira. Compounded, naira costs rose about 78% and dollar costs 25%. The exchange rate explains little of this: at first-quarter rates the naira fell about 9%, then rose about 11%, ending almost where it started. The rest comes from changes in the survey's own Lagos rates. Nairobi moved from US$785 to US$851.

Between 2019 and 2025 the euro-linked CFA francs and the dirham gained about 4 to 5% against the dollar, the Kenyan shilling lost about 21% and the naira went from 361 to 1,450. In 2025 most currencies strengthened: the Congolese franc by 28%, the CFA franc by 13%, the dirham by 11%, and the Egyptian pound and the naira by 6 to 7%. Dollar comparisons need the same year and rate date.

Labour is cheap in dollars almost everywhere. Statutory minimum wages run from about US$48 a month in Nigeria to US$354 in Morocco. Kenyan construction workers earned KES29,876 to 53,629 a month in 2025 (US$231 to 415), and Egypt's CAPMAS put the average construction wage at EGP7,340 a month in 2023 (about US$240), above the all-sector average. Kenyan construction pay fell about 10% in real terms between 2021 and 2025. At 21.45% of Kenya's index, a 10% wage rise adds about 2% to direct cost.

+78%

Lagos construction costs in naira across two survey years; +25% in dollars

US$48-354

monthly statutory minimum wage, Nigeria to Morocco, at end-2025 and 2026 rates

+6.7%

the Egyptian pound against the dollar in 2025

CHAPTER 5



05 When building technology pays

Precast concrete, light steel frames, insulated panels, volumetric modules and 3D printing all promise to build faster and cheaper. The record in Africa and the Gulf is long on capacity and short on audited cost.

Building technology is usually sold on cost. Dubai's 3D printing strategy stated that printing would cut construction costs by 50 to 70%. The completed African projects report far less: VOA News reported that 14Trees' 3D-printed school in Malawi cost about 15% less to print than to build. No African 3D-printed home, panel house or precast scheme in our research publishes an outturn cost per m^2 against a conventional comparator.

Time is a different matter. Walls printed in 12 to 18 hours, a 64 m^2 house finished in 14 days, a precast floor a week instead of a month and light-steel-frame time savings of over 30% recur across sources. Chapter 4 showed what time is worth: in a city where money costs 20% or more, a faster programme is a direct cost saving.

Volume decides whether either saving appears. Saudi Arabia reports modern methods in 58% of its housing system after pairing guaranteed state demand with factory finance. Kenya's state panel plant has published capacity but no output, and Ethiopia's precast factory had never run above two-thirds of its design by 1998.

This chapter grades the claims, sets out the conditions under which a factory pays, and measures what speed is worth in each city.

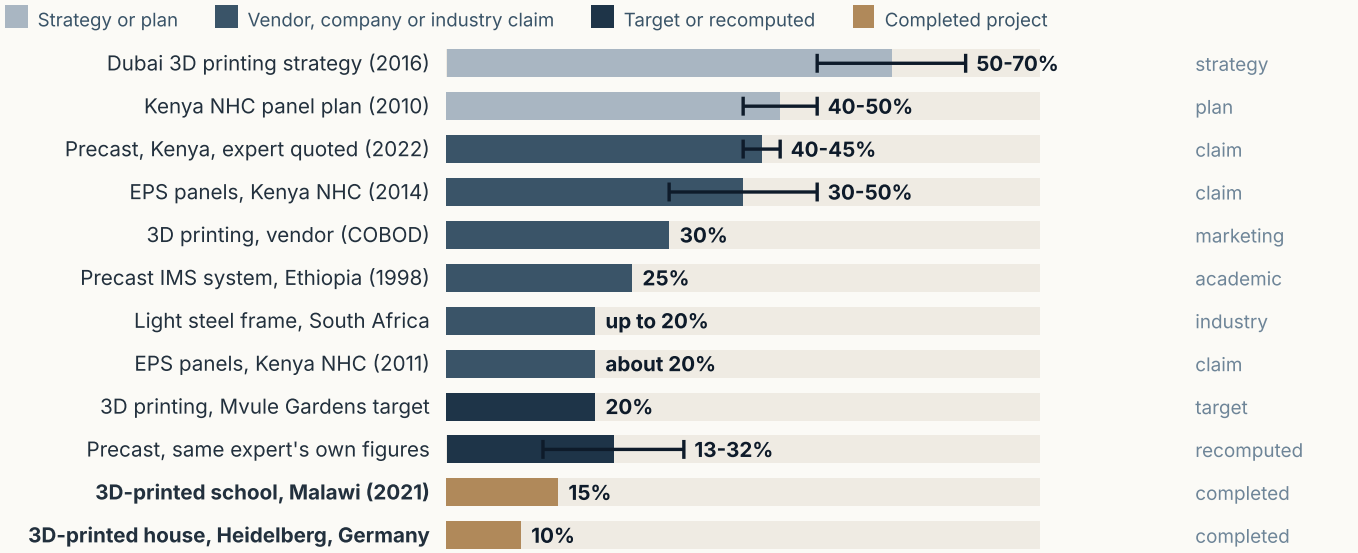
THE EVIDENCE

Claimed savings shrink as the evidence firms up; time savings hold

EXHIBIT 17

Cost savings fall from 50 to 70% in Dubai's 3D printing strategy to 10 to 15% on completed projects

Stated construction cost saving against conventional building, %, by type of evidence; lines show stated ranges



Source: Dubai 3D Printing Strategy via Arch2O; Business Daily (2010); The Star, Kenya (2022); Construction Kenya (2014); Capital FM (2011); COBOD (2026); Addis Ababa University thesis (1998, citing Yugoslav data); SASFA via BuildSteel (2017); COBOD project reports (2023); VOA News (2021). **Heirstone analysis** recomputes the Kenyan precast expert's own figures (KSh2.6m against KSh3.0 to 3.8m for a three-bedroom house).

We ranked every cost claim in our research by how close it stands to a measured outturn. Strategy documents and plans sit at the top, at 40 to 70%. Vendor and company claims sit at 20 to 50%, and they vary over time: Kenya's National Housing Corporation described its panel savings as up to 40% or half in 2010, about 20% when its factory opened in 2011, and up to 30% in 2014, in an article that also called them nearly half. Targets come next. Completed projects sit at the bottom: about 15% on the Malawi school, as reported by VOA News, and 10% on a 3D-printed house in Heidelberg, published by the printer's maker.

Figures within one source can differ. An expert quoted in Kenya's press in 2022 put precast savings at 40 to 45% on a two-bedroom house, while the same expert's three-bedroom figures, KSh2.6m precast against KSh3.0 to 3.8m conventional, imply 13 to 32%. 14Trees itself said in 2023 that printing already cost the same as conventional building and aimed to bring it 20% lower.

Government targets sit between claims and outturns. Dubai's 2021 decree supports a goal of 3D printing in 25% of the emirate's buildings by 2030, and the municipality issued its first construction licence for a privately owned 3D-printed villa in 2023. We found no published tally of progress towards the goal.

TIME SAVINGS CLAIMED OR REPORTED

Dubai strategy (strategy)	**90%**
COBOD marketing (vendor)	**50%**
Amana / DuBox modules (company)	**30-50%**
CIMC hotel modules, Riyadh (projected)	**over 40%**
Precast floor, Kenya (company)	**a week, not a month**
Light steel frame, 1,800 m² (completed)	**over 30%**
PERI, Heidelberg (completed)	**30%**
Echostone, Lagos (prototype)	**14 days a house**

Sources as Exhibit 17; Technical Review Middle East (2023); Hotel Online (2026); Engineering News (2013); The Guardian, Nigeria (2018).

THE HEIRSTONE VOLUME TEST

Five conditions decide whether a building factory pays. The two programmes that reached scale meet at least four.

A factory turns a fixed cost into a cheaper or faster unit only if it runs near capacity, and it runs near capacity only if five conditions hold together. We drew them from the cases in our research and scored each case against them.

Saudi Arabia meets the first four: state demand, factories sized to it, approved techniques and finance for both. Egypt meets all five by a different route. It builds at very large volume with cement and steel made at home, and adds precast where it pays: Arab Contractors runs a 50,000 m² a year hollow-core slab plant, and Orascom Construction holds an exclusive partnership with COBOD for 3D printing.

Where one condition fails, output falls short. Ethiopia's plant had state backing and a site 10 km from the capital, yet peaked at 66% of its design. Mvule Gardens has finance and certification, but 52 homes are a demonstration within a national programme of 262,913 units under construction.

Pipeline	A committed buyer for repeat units over several years. Saudi Arabia's first funded factories were sized at 624 to 1,500 units a year each; Kenya's state developer said in 2011 that its output of about 100 houses a year was not enough.
Volume	Output close to design capacity, measured and published. Capacity announcements are not output.
Approvals	An approved standard for the system. Saudi Arabia recognises six modern building techniques; a Kenyan trade report noted that EPS panels had not been standardised.
Finance	Capital for the factory and mortgages or offtake for the buyers. Saudi Arabia financed four factories in 2022; Kenya funds its programme with a 1.5% payroll levy.
Distance	A factory within trucking distance of the sites. Ethiopia's plant was 10 km from Addis Ababa; a printer prints on site.

EXHIBIT 18

Saudi Arabia and Egypt pass; the factories that fell short missed pipeline or volume

Heirstone Volume Test applied to industrialised and large-scale building programmes in our research

● met ◑ partly met ○ not met n/p not published

CASE	COUNTRY	PIPELINE	VOLUME	APPROVALS	FINANCE	DISTANCE	WHAT HAPPENED
Saudi factory-built housing	Saudi Arabia	●	●	●	●	n/p	Modern methods in 58% of the housing system; 22+ factories, over 40,000 units a year of capacity
Public housing at volume, with selective precast	Egypt	●	●	●	●	●	685,000 finished social units by mid-2024; local cement and steel; 50,000 m² a year of hollow-core slabs
Precast sector	Morocco	●	◑	n/p	●	●	Precast takes 9.5% of cement deliveries; 300 firms; MAD1,134m of new plant
Echostone with Lagos State	Nigeria	●	◑	●	n/p	●	252 flats finished by end-2019 of a 2,000-unit first phase
NHC panel factory, Mavoko	Kenya	◑	n/p	◑	●	●	Capacity of 126,720 panels a year; output not published
PBPPE precast plant, Kaliti	Ethiopia	◑	○	●	●	●	Peaked at 66% of design; survey respondents cited awareness, sizes and looks
Mvule Gardens, 3D printing	Kenya	○	○	●	●	●	52 homes, about 0.02% of the national programme under construction

Source: Saudipedia (2023); CAHF (2024); The Arab Contractors (2026); Orascom Construction (2022); APC via CemNet (2026); EcoActu (2022); InvestData (2019); Lagos HOMS; Capital FM (2011); Construction Kenya (2014 and undated); Addis Ababa University thesis (1998); COBOD (2023); 14Trees (2026); Capital FM via AllAfrica (2026). **Heirstone analysis.**

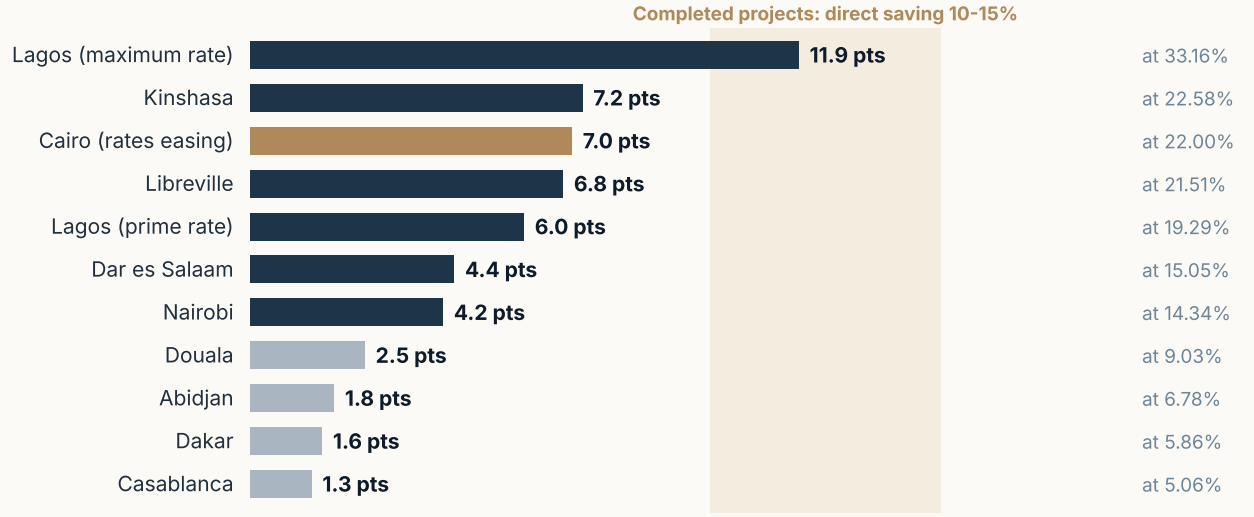
WHAT SPEED IS WORTH

Speed is worth most where money is dear: a quarter off the programme saves 6 to 12 points in Lagos and 1.3 in Casablanca

EXHIBIT 19

Finishing in 18 months rather than 24 saves 4 to 12 points of hard cost in interest in five of the nine cities, against a 10 to 15% direct saving on completed projects

Interest saved by cutting a fully debt-funded build from 24 to 18 months, percentage points of hard cost, at the latest average lending rate



Source: As Exhibit 13; completed-project savings from Exhibit 17. **Heirstone analysis.**

The time claims in the previous pages cluster at 30 to 50%. We take a more cautious quarter, a 24-month programme finished in 18, and value it at each city's lending rate. In Lagos, Kinshasa and Libreville, that quarter is worth 6 to 12 points of hard cost in interest alone, half or more of the 10 to 15% direct saving reported on completed industrialised projects. In Cairo it is worth about 7 points and shrinks with each cut in rates. In Dar es Salaam and Nairobi it is worth about 4 points; in Kenya, where houses take 3.3 years, cutting a three-year build to two is worth 9. In Douala, Abidjan, Dakar and Casablanca, where credit is cheap, it is worth 1.3 to 2.5 points, and a faster method has to pay on direct cost.

Volume is the constraint. One printer built about one house a week at Mvule Gardens. Kenya's target of 200,000 affordable units a year would need about 3,800 printers at that rate, roughly US$1.9bn at the US$500,000 a printer costs, as reported by VOA News. Saudi Arabia's factories, sized at hundreds to thousands of units a year, are the scale at which industrialisation changes a national cost base.

Green certification is a finance lever more than a construction method. Africa's EDGE-certified floor space rose from 4.6 million m^2 in 2023 to about 10 million in 2025. Kenya passed 1 million m^2 across 212 buildings, and Egypt is close to 1 million, including the Grand Egyptian Museum, certified with over 60% lower energy costs and 34% less water.

3,800

printers needed for 200,000 homes a year at one house a week each (Heirstone analysis)

0.02%

Mvule Gardens' 52 homes as a share of Kenya's 262,913 programme units under construction

~1m m^2

EDGE-certified floor space in Egypt, 2025



CHAPTER 6

06 From Casablanca to Dar es Salaam

North Africa: Egypt and Morocco

Egypt is the reference point for the sample because it combines cheap structural materials made at home with a long record of building at very large volume, and its money is getting cheaper. Its cement industry produced about 65 Mt in 2025 and exported 18.5 Mt, within a cap of 30% of production, and its steel industry makes more than a third of Africa's crude steel. The Social Housing Program announced in 2014 delivered 296,022 units by the end of 2019, about 59,000 a year. The building code caps permit processing at 30 days. Knight Frank values Egypt's project pipeline at over US$565bn, with about US$120bn under construction, and Turner & Townsend names Cairo, with Doha and Riyadh, as a market with many projects coming to tender in 2026.

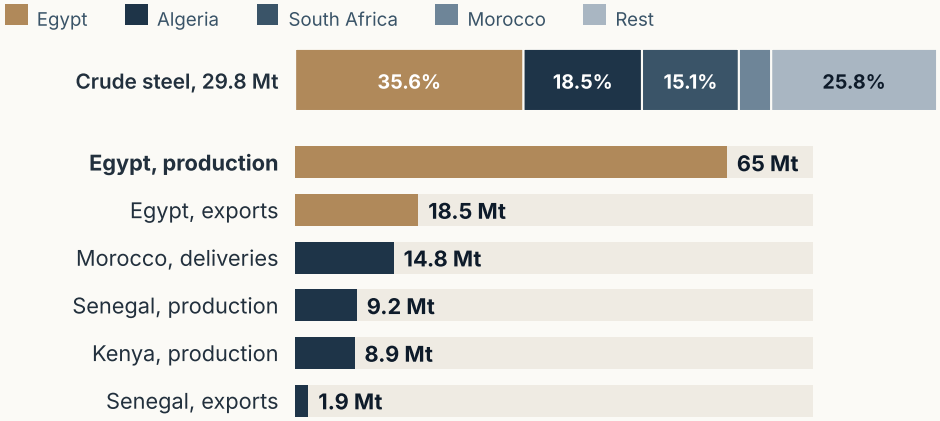
Morocco has cheap money and a deep cement industry, but no published construction cost. Cement deliveries reached 14.8 Mt in 2025, up 8.2%, about 70% of it for housing. Precast products took 9.5% of deliveries in the first nine months of 2026 and ready-mixed concrete 26.8%, a baseline for industrialised building that no other country in the sample publishes. Real-estate loans averaged 5.06% in the second quarter of 2026. The nearest public benchmark for Casablanca is a social unit sold at MAD250,000 for 50 m^2, about US$503 per m^2.

For the cities south of the Sahara, Egypt also matters as a supplier. Its cement and clinker already reach Côte d'Ivoire and Ghana, Titan Egypt is building export capacity at Alexandria, and Egyptian rebar trades close to the Turkish export price.

EXHIBIT 20

Egypt makes more than a third of Africa's crude steel, and its cement and clinker exports alone exceeded Senegal's entire output in 2025

Crude steel output by country, Africa, 2025; cement production, deliveries and exports, million tonnes, 2024 or 2025



Source: worldsteel, World Steel in Figures 2026; CemNet (2026); APC via Le Matin (2026); Senegal Ministry of Planned Economy via CemNet (2026); Kenya Economic Survey via Global Cement (2025, figures for 2024). **Heirstone analysis** for shares.

WEST AND CENTRAL AFRICA: THE CFA FRANC CITIES AND KINSHASA

Five francophone cities, two currency unions and one common gap: no current cost guide prices them

Abidjan and Dakar borrow on the West African union's terms: a 3.00% policy rate, bank lending between about 6 and 7%, and inflation close to zero. Interest on a two-year build is about 6 to 7% of hard cost, so the first lever is cement. Côte d'Ivoire has more capacity than it uses, on estimates published by two business websites, regulated ex-factory ceilings of CFA72,000 to 78,000 a tonne and a 3 Mt grinding plant opened at Attinguié in 2025, but it still grinds largely imported clinker. Senegal is a net exporter: it produced 9.247 Mt in 2025 and exported 1.931 Mt, and the new Sococim kiln removes up to 400 kt of clinker imports a year. At CFA3,500 a bag, Dakar's cement is the cheapest of the CFA franc cities in our sample.

Douala and Libreville share the BEAC's terms but not its credit. Cameroon's average lending rate, 9.03%, is low by the standards of the sample; Gabon's, 21.51%, is among the highest. In Cameroon, cement is the binding cost. Capacity of 10 to 12.7 Mt against about 8 Mt of demand has not removed the dependence on imported clinker, which Figuil's new clinker plant and Kribi's low-clinker cement begin to address, and regulated prices in Douala were set at CFA4,450 to 5,000 a bag at the end of 2023. In Gabon, finance and distance matter most. CIMAF's Owendo plant has 0.85 Mt of capacity for a market of about 0.9 Mt; the official price list adds up to 60% for interior towns; and a Libreville builder put the labour cost of a house at CFA80,000 per m^2 (US$131).

Kinshasa is the dearest cement market in the sample at US$11.03 a bag, and the most exposed to the road from Kongo Central. National capacity of about 4.64 Mt is set to grow, with CIMKO planning 2.91 Mt by 2027 at a cost of about US$300m and CILU 2.2 Mt. The Congolese economy is dollarised: about 95% of bank loans are in US dollars, so the franc lending rate of 22.58% overstates what most developers pay, and the cheapest new formal house, US$25,763 for 100 m^2 in 2022, is priced in dollars too.

All five cities share one gap. None appears in a current international cost guide, and the averages that circulate are mostly prestige-office figures. A city-level construction cost survey, published each year by a statistics office, a central bank or a professional body, would help every lender and investor that prices a project there. Côte d'Ivoire, Senegal and Ghana together already have nearly 2 million m^2 of EDGE-certified floor space, so the projects exist to measure.

3.00%

BCEAO policy rate since early 2026; WAEMU real-estate loans averaged 6.67% in 2025

CFA3,500

a bag of cement in Dakar (US$5.73), the lowest in the CFA franc zone in our sample

21.51%

average bank lending rate in Gabon; Cameroon 9.03%, same policy rate

95%

of bank loans in DR Congo are in US dollars

0

francophone cities in the sample with a current construction cost guide rate

LAGOS, NAIROBI AND DAR ES SALAAM

The three cities with the best cost data face two different problems: credit in Lagos, the customs band in Nairobi and Dar es Salaam

Lagos is the dearest city in the published guides and the one where materials explain least. Nigeria has two to three times the cement capacity it consumes, and Dangote Cement plans to reach 80 Mt a year across Africa by 2030, with a 6 Mt plant at Itori due by the end of 2026. Yet credit and margin dominate. Prime borrowers paid an average of 19.29% early in 2026, and the average maximum rate reached 33.16% by mid-year, so the gap between the best and the riskiest borrowers is worth 18 points of hard cost on a two-year build, more than any cement saving. The central bank cut its policy rate from 26.5% to 23% later in 2026. Turner & Townsend forecasts 18% cost escalation in Lagos in 2026, the highest in its Africa sample.

Nairobi is the cheapest city in Turner & Townsend's survey and the cheapest for housing in AECOM's, and its costs barely moved in 2025: Kenya's construction input index rose 0.74% in the year to the third quarter, before a 5.73% quarterly rise in 2026 that the press linked to fuel. Its largest lever is the border: the East African Community's 35% band on cement and rebar and Kenya's 17.5% levy on imported clinker and steel inputs. Time matters too: a house took 3.3 years on average in 2023. Kenya also runs a large test of industrialised building, with 262,913 units under construction in its affordable housing programme, funded by a 1.5% payroll levy.

Dar es Salaam prices close to Nairobi but imports more. CAHF reports that 70% of essential items such as cement and steel come from abroad, the EAC tariff applies, and the overall lending rate was 15.05% in 2026. Supply has been uneven: a 2026 price survey averaged TZS321,251 a tonne before VAT while some outlets sold bags at up to TZS26,000. Tanzania's 2026 wage order sets construction minimum wages at TZS398,500 to 515,000 a month (US$151 to 195).

The Gulf enters the story as a reference and as a source of capacity. Dubai builds at US$1,990 per m^2 and Riyadh at US$2,874 on Turner & Townsend's basket, and Saudi Arabia's factories offer the clearest model of state-backed industrialisation. Gulf module plants such as Amana's at Rabigh, which can make 16 modules a day, and a planned NEOM factory sized for 12,500 a year, show the scale a factory reaches with a pipeline behind it.

18.4 pts

of hard cost: the gap between Nigeria's prime and maximum lending rates on a two-year build

14.1

Nairobi on Turner & Townsend's index (London = 100), the lowest in Africa and the Middle East

70%

of essential building items imported in Tanzania, according to CAHF

262,913

units under construction in Kenya's affordable housing programme, 2026

CHAPTER 7

07 The Heirstone Cost Ladder

The ladder measures five levers on one scale, percentage points of construction cost, for each city, using only figures shown in this report and Kenya's input weights as a common mix. The largest lever is the top rung: the place to start.

1	**Speed**	Interest saved by finishing a fully debt-funded build in 18 months rather than 24, at the city's average lending rate. Industrialised building and better programming deliver it.
2	**Credit**	Interest saved by borrowing at the best published rate in the same market or currency union: prime rather than maximum in Lagos, Cameroon's average in Libreville.
3	**Margin**	Overheads, profit and preliminaries saved by letting one large package rather than small ones. Published for Lagos and Nairobi only.
4	**Border**	The most that removing duty on cement and rebar could save if prices sit at import parity.
5	**Cement**	The saving if a bag cost what it costs in Egypt. Clinker made at home, shorter haulage and regional sourcing are the routes.

EXHIBIT 21

The first lever differs by city: credit in Lagos and Libreville, the border in East Africa, cement and haulage in Kinshasa, Douala, Abidjan and Casablanca

Heirstone Cost Ladder: points of construction cost each lever could save; gold is the top rung, sand the second. Levers overlap and are not additive

CITY	SPEED 24 TO 18 MONTHS	CREDIT BEST RATE IN MARKET	MARGIN SMALL TO LARGE PACKAGE	BORDER DUTY ON CEMENT AND REBAR	CEMENT GAP TO CAIRO PRICE	TOP RUNG
Lagos	**6.0-11.9**	**18.4**	7.9	1.8*	7.3	**Credit**
Libreville	**6.8**	**14.4**	*n/p*	4.1	6.2	**Credit**
Nairobi	4.2	*n/p*	**4.4**	**6.4**	4.1	**Border**
Dar es Salaam	**4.4**	3.5	*n/p*	**6.4**	3.5	**Border**
Kinshasa	**7.2**	*n/p*	*n/p*	3.1	**8.9**	**Cement and haulage**
Douala	**2.5**	1.5	*n/p*	1.8*	**7.2**	**Cement and clinker**
Abidjan	**1.8**	1.0	*n/p*	1.8*	**7.0**	**Cement and clinker**
Casablanca	1.3	0.3	*n/p*	**1.9***	**6.9**	**Cement**
Dakar	1.6	0.0	*n/p*	**1.8***	**4.0**	**Short ladder**
Cairo	**7.0**	*cut by 10.9 since 2024*	*n/p*	0.0	0.0	**Speed**

Source: Heirstone analysis from Exhibits 6, 7, 11, 13, 14 and 15 and their sources. * Rebar only: Nigeria, Cameroon, Côte d'Ivoire, Senegal and Morocco have more cement capacity than they use, so local cement prices sit below import parity. Cairo is the benchmark for the cement rung and charges no duty on cement or rebar; its central bank's easing has already cut interest on a two-year build by 10.9 points.

The ladder sorts the nine cities into four groups. Credit comes first in Lagos and Libreville, at 18.4 and 14.4 points, more than any materials saving. The border comes first in Nairobi and Dar es Salaam, at 6.4 points, because the EAC's 35% band covers both cement and rebar. Cement comes first in Kinshasa, Douala, Abidjan and Casablanca, at 6.9 to 8.9 points: in Kinshasa that means the road, in Douala and Abidjan imported clinker. Dakar has the shortest ladder, with no lever above about 4 points. Cairo has already climbed the materials rungs, and its remaining rung, speed, is worth 7 points and falling as rates ease. Technology enters through speed, so it ranks high in Lagos, Kinshasa and Libreville, where the Volume Test decides whether a factory can deliver it.

WHAT THE LADDER LEAVES OUT

Labour

Statutory wages run from about US$48 to US$354 a month; a 10% wage change moves direct cost by about 2%.

Specification

In Douala the published average is 5.9 times the cost of state social housing; the brief moves cost more than any rung.

Currency

Lagos's dollar cost fell 23%, then rose 63%, between surveys; the ladder is measured in local terms.

WHAT TO DO

Start with the largest rung in the city, and treat technology as a way to buy time

The levers that would cheapen a square metre differ from city to city, and most of them sit with lenders, customs administrations and procurement teams rather than with the people who choose the building system.

FOR

Governments, land agencies and investment promotion agencies

Start with the top rung in your own city. In the EAC, review the 35% band and the clinker and steel levies on structural inputs for housing; in Gabon and the DRC, the haulage corridor from plant to city; in Cameroon, Gabon and Côte d'Ivoire, clinker made at home.

Publish a construction cost index by city and building type, with its weights, as Kenya does. Morocco and the CFA franc cities have none.

Approve standards for industrialised systems before funding factories, and cap permit times; Egypt's code sets 30 days.

Publish official cement prices and haulage surcharges by town, as Gabon and Côte d'Ivoire do, so that buyers can see what distance costs.

FOR

Lenders, DFIs, pension funds and asset managers

Cheaper and longer construction credit is the largest lever in Lagos and Libreville, worth 14 to 18 points of hard cost on a two-year build, more than any material.

Underwrite programme length as closely as the rate. A third year adds 9 points of interest in Nairobi and 16 in Kinshasa.

Lend to building factories only against a committed pipeline that passes the Volume Test, and to buyers through green and EDGE-linked products.

Focus long-term lending on Central Africa, where long-term loans are under 5% of new bank credit in Cameroon and Gabon.

FOR

Developers, contractors and corporate occupiers

Package work at scale: in Lagos a large package cuts about 8% from overheads, profit and preliminaries.

Source structure regionally. Egyptian rebar trades near the Turkish export price, and Egyptian cement already reaches Côte d'Ivoire and Ghana.

Budget in local currency with the exchange-rate date stated, and choose a building system by the city's top rung: speed where money is dear, material efficiency where it is cheap.

Where a city has only a prestige-office average (grade C in the index), commission a local cost plan before fixing a budget.

How Heirstone works on construction cost

We help developers, lenders, investors and governments in the Middle East and Africa decide what to build and what it will cost: market and demand studies, highest-and-best-use and site selection, feasibility and financial models (DCF, IRR, NPV and Monte Carlo testing), cost and programme benchmarking of the kind in this report, and the case for or against a building factory or a housing programme.

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Cost per m²: construction cost per m² of gross floor area, excluding land, professional fees, VAT and finance unless stated. **Hard cost:** the contract sum, including preliminaries and contractor overheads and profit. **Preliminaries:** the contractor's site running costs. **Clinker:** the kiln product ground with gypsum and additives to make cement. **CIF:** landed value of an import, including cost, insurance and freight. **Prime and maximum rates:** the average rates Nigerian banks charge their best and their riskiest borrowers.

Method

Costs are as published, in the currency, year and exchange rate of each source; we do not restate earlier years in current prices. Interest during construction assumes equal monthly drawdowns, monthly compounding and interest rolled up to completion: IDC/C = $[((1+r/12)^n - 1)/(r/12) - n]/n$. For Cairo the rate is a Heirstone assumption: the Central Bank of Egypt's overnight lending rate plus 2 points, the margin CAHF reports for Egyptian mortgages. Heirstone estimates for multi-storey housing apply 0.72, the median ratio of AECOM's housing to prestige-office rates in Lagos, Accra, Johannesburg, Dar es Salaam and Nairobi (range 0.43 to 0.82), to each city's published average. Cost Ladder rungs use Kenya's building cost index weights (cement and lime 14.18%, steel 10.69%) as a common input mix. The duty and VAT wedge is CIF x (1 + duty) x (1 + VAT), other levies excluded. Exchange rates are US Treasury reporting rates or World Bank annual averages unless a source states its own.

Exchange rates

UNITS PER US$	CAHF, MID-2024	2025 AVERAGE	MARKET, 2026
Nigerian naira	1,529	1,518.4	1,331.3
Kenyan shilling	128.99	129.30	129.63
Tanzanian shilling	2,649.7	2,537.6	2,633.8
CFA franc (XOF / XAF)	610.5 / 610.0	581.9	586.1
Congolese franc	2,864.1	2,693.2	2,311.6
Moroccan dirham	9.95	9.35	9.77
Egyptian pound	48.23	49.23	52.80

2025 averages: World Bank official rates; Tanzania from the Bank of Tanzania. Market 2026: ExchangeRate-API, latest available. Other conversions use US Treasury reporting rates for the quarter stated by the source.

Use of this report

This report is published for general information. It is not investment, legal, valuation or engineering advice and should not be relied on as such. Figures from third parties are reproduced as published and have not been audited by Heirstone Consulting. Images are illustrative.

1 WHAT A SQUARE METRE COSTS

Turner & Townsend, International Construction Market Survey 2024 and Global Construction Market Intelligence 2025 and 2026 (Africa, Middle East, UK and methodology pages; published market metrics and cost data sheets); AECOM, Africa Property & Construction Cost Guide 2024/25; Rider Levett Bucknall, The Cost of Building in Africa (2022); Knight Frank, Egypt construction market release (2025) and EnterpriseAM (2025); CAHF, Housing Finance in Africa Yearbook 2024, country profiles; Kenyans.co.ke (2024); Creamer Media (2025); TelQuel and MAP citing FNPI (2026); GeoNames.

2 CEMENT AND STEEL

KNBS, Construction Input Price Indices (2025) and Capital FM and Business Daily reporting of KNBS data (2025, 2026); Dangote Cement FY2025 earnings release and results presentation; FCCPC via CemNet (2026); CemNet and International Cement Review (2025, 2026); Global Cement citing the Federation of Egyptian Industries and the Export Council for Building Materials (2025); S&P Global Platts (2025); Arab Finance via Zawya (2026); Daba Finance (2025); Erooamba (2026); Heidelberg Materials (2026); worldsteel, World Steel in Figures 2026; Arab Iron and Steel Union price reports (2025, 2026); Fraternité Matin (2025); EcoMatin (2023); Business in Cameroon (2025); Holcim Maroc and Ciments du Maroc annual reports (2025).

3 DUTY, LEVY AND HAULAGE

World Bank WITS and UNCTAD TRAINS applied tariffs; East African Community (2022); PwC Worldwide Tax Summaries (2026); FERDI (2014); Business Daily (2025); The Standard (2026); Vicat FY2025 results (2026); Government of Gabon price order via CemNet and L'Union (2023); Afrique-sur7 (2025); AllAfrica (2024); US Treasury reporting rates of exchange.

4 THE PRICE OF TIME

Central banks: BCC, BEAC (including its first-quarter 2026 report on lending rates), BCEAO, CBN, CBK, Bank of Tanzania, Bank Al-Maghrib, Central Bank of Egypt and ECB, directly or as reported by ThisDay, Arise News, Challenge.ma, FocusEconomics, Amwal Al Ghad, Egypt Today, Gabon Media Time, EcoMatin and Financial Afrik (2025, 2026); World Bank WDI and IMF WEO; CAPMAS (2024); KNBS Economic Survey 2026 via Kenyan Wallstreet; WageIndicator; Clyde & Co (2026); State House Nigeria (2024); Al-Ahram Weekly (2025); Bladi.net (2025).

5 BUILDING TECHNOLOGY

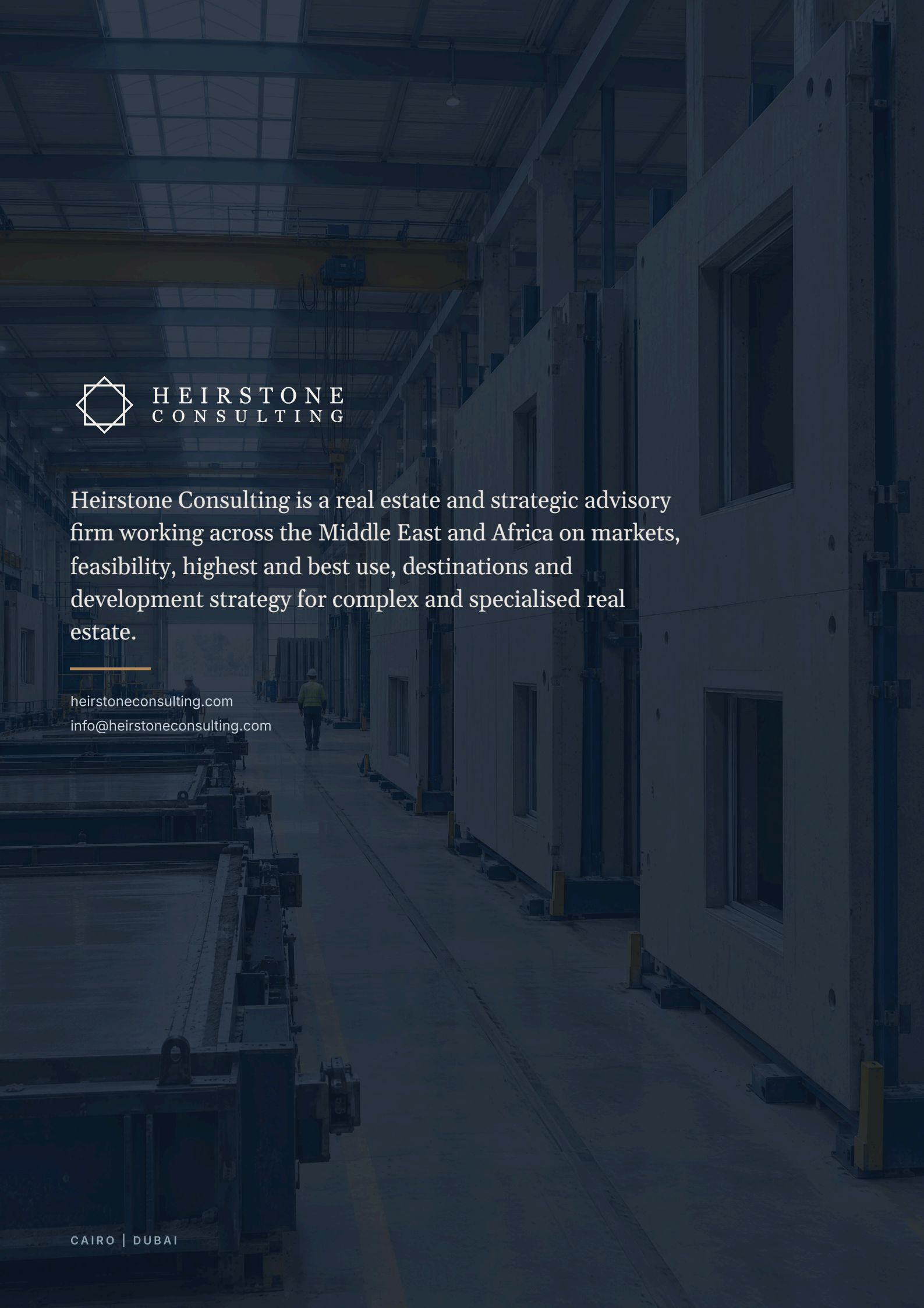
COBOD (2023, 2026); 14Trees (2026); EDGE Buildings, IFC (2023, 2025); VOA News (2021); African Business (2023); Engineering News and Mining Weekly (2013, 2021); BuildSteel (2017); Khaleej Times (2021); Arch2O; Gulf News (2023); MEED (2022); Orascom Construction (2022); The Arab Contractors (2026); Business Daily (2010); Capital FM (2011); Construction Kenya; Kenya Yearbook (2022); The Star, Kenya (2022); Kenya News Agency (2025); Addis Ababa University thesis on PBPPE (1998); UN-Habitat (2010); The Guardian, Nigeria (2018); InvestData (2019); Lagos HOMS; Saudipedia (2023); Construction Dive (2021); Technical Review Middle East (2023); Hotel Online (2026); EcoActu (2022); Structurama (2019); The Standard, Kenya (2025); London South Bank University paper on Egyptian social housing.

6 AND 7 REGION AND COST LADDER

Sources for chapters 2 to 5; Knight Frank (2025); Turner & Townsend (2026); CAHF Housing Finance in Africa Yearbook 2024 (Egypt, Morocco, Kenya, Tanzania, DRC, Gabon and Cameroon profiles); APC via Le Matin and CemNet (2026); Ciments du Maroc (2026); Bladi.net (2025); Erooamba (2026); CemNet on DR Congo, Gabon, Cameroon and Tanzania (2025, 2026); Dangote Cement Capital Markets Day via CemNet (2026); Business Daily (2026); Capital FM via AllAfrica (2026); Clyde & Co (2026).

What the public record does not show

No public construction cost per m² for Casablanca and no current city rates for the five francophone cities south of the Sahara; no published haulage cost per tonne of building material in any of the ten countries; no audited outturn cost per m² for any African industrialised housing scheme; no recent average commercial lending rate for Egypt.



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